

Easy Visa Business Report

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Index

1. Exploratory Data Analysis (8 Points)

1.1 Problem Definition

1.2 Objective

1.3 Data Description and Initial Observations

1.4 Univariate Analysis

1.4.1 Continent

1.4.2 Education of Employee

1.4.3 Job Experience

1.4.4 Job Training Requirement

1.4.5 Number of Employees

1.4.6 Year of Establishment and Employer Age

1.4.7 Region of Employment

1.4.8 Prevailing Wage

1.4.9 Unit of Wage

1.4.10 Full-Time Position

1.4.11 Case Status Distribution

1.5 Bivariate Analysis

1.5.1 Correlation among Numerical Variables

1.5.2 Prevailing Wage by Case Status

1.5.3 Employer Age by Case Status

1.5.4 Number of Employees by Case Status

1.5.5 Case Status by Education Level

1.5.6 Case Status by Continent

1.5.7 Case Status by Job Experience

1.5.8 Case Status by Job Training Requirement

1.5.9 Case Status by Region of Employment

1.5.10 Case Status by Unit of Wage

1.5.11 Case Status by Full-Time Position

1.6 Key Insights from EDA

– Patterns and meaningful observations from univariate and bivariate analysis

2. Data Preprocessing (5 Points)

2.1 Data Quality Checks

- Missing Value Treatment
- Duplicate Records Check
- Handling Negative or Impossible Values

2.2 Outlier Detection

2.3 Outlier Treatment

- Log Transformation
- Capping (IQR Method)

2.4 Feature Engineering

- Employer Age Derivation
- One-Hot Encoding of Categorical Variables
- Target Encoding and Label Transformation

2.5 Data Splitting Strategy

- Train, Validation, and Test Split
 - Ensuring No Data Leakage
-

3. Model Building – Original Data (6 Points)

3.1 Model Selection and Evaluation Metric

- Rationale for Choosing Recall

3.2 Models Built

- Decision Tree
- Bagging
- Random Forest
- AdaBoost
- Gradient Boosting

3.3 Model Evaluation

- Cross-Validation Results
- Validation Results

3.4 Interpretation and Key Observations

4. Model Building – Oversampled Data (6 Points)

4.1 Rationale for Oversampling (SMOTE)

4.2 Models Trained

- Decision Tree

- Bagging
- Random Forest
- AdaBoost
- Gradient Boosting

4.3 Cross-Validation and Validation Results

4.4 Insights and Interpretation

5. Model Building – Undersampled Data (6 Points)

5.1 Rationale for Undersampling

5.2 Models Trained

- Decision Tree
- Bagging
- Random Forest
- AdaBoost
- Gradient Boosting

5.3 Cross-Validation and Validation Results

5.4 Key Observations

6. Model Performance Improvement using Hyperparameter Tuning (13 Points)

6.1 Selection of Models for Tuning

6.2 Tuning Approach (Randomized SearchCV)

6.3 Tuned Models and Best Parameters

- AdaBoost (Undersampled Data)
- AdaBoost (Original Data)
- Gradient Boosting (Undersampled Data)
- Gradient Boosting (Original Data)

6.4 Comparative Performance Analysis

6.5 Model Interpretations and Conclusions

7. Model Performance Comparison and Final Model Selection (4 Points)

7.1 Training Performance Comparison

7.2 Validation Performance Comparison

7.3 Final Model Selection

7.4 Feature Importance Analysis

7.5 Key Takeaways

8. Actionable Insights & Recommendations (8 Points)

8.1 Summary of Analytical Insights

8.2 Business Recommendations

8.3 Policy and Strategic Implications

List of Figures

- Figure 1:** Data Overview
- Figure 2:** Statistical Summary
- Figure 3:** Barplot and Frequency Table of *Continent*
- Figure 4:** Barplot and Frequency Table of *Education of Employee*
- Figure 5:** Barplot and Frequency Table of *Has Job Experience*
- Figure 6:** Barplot and Frequency Table of *Requires Job Training*
- Figure 7:** Histogram and Boxplot of *Number of Employees*
- Figure 8:** Histogram and Boxplot of *Year of Establishment*
- Figure 9:** Histogram and Boxplot of *Employer Age*
- Figure 10:** Barplot and Frequency Table of *Region of Employment*
- Figure 11:** Histogram and Boxplot of *Prevailing Wage*
- Figure 12:** Barplot and Frequency Table of *Unit of Wage*
- Figure 13:** Barplot and Frequency Table of *Full-Time Position*
- Figure 14:** Barplot and Frequency Table of *Case Status*
- Figure 15:** Heatmap for Numerical Variables
- Figure 16:** Pair Plot for Numerical Variables
- Figure 17:** Boxplot of *Prevailing Wage by Case Status*
- Figure 18:** Boxplot of *Employer Age by Case Status*
- Figure 19:** Boxplot of *Number of Employees by Case Status*
- Figure 20:** Barplot of *Case Status by Education Level*
- Figure 21:** Barplot of *Case Status by Continent*
- Figure 22:** Barplot of *Case Status by Job Experience*
- Figure 23:** Barplot of *Case Status by Requirement of Job Training*
- Figure 24:** Barplot of *Case Status by Region of Employment*
- Figure 25:** Barplot of *Case Status by Unit of Wage*
- Figure 26:** Barplot of *Case Status by Full-Time Position*
- Figure 27:** Missing Value Analysis
- Figure 28:** Detection of Impossible Negative Values in *No. of Employees*
- Figure 29:** Outlier Detection for Numerical Variables
- Figure 30:** Outlier Treatment (Log Transformation Results)
- Figure 31:** Outlier Treatment – *Employer Age*
- Figure 32:** Algorithm Comparison (Cross-Validation Recall) on Original Data
- Figure 33:** Algorithm Comparison (Cross-Validation Recall) on

Oversampled Data

Figure 34: Algorithm Comparison (Cross-Validation Recall) on Undersampled Data

Figure 35: Model Performance – *Tuned AdaBoost (Undersampled Data)*

Figure 36: Model Performance – *Tuned AdaBoost (Original Data)*

Figure 37: Model Performance – *Tuned Gradient Boosting (Undersampled Data)*

Figure 38: Model Performance – *Tuned Gradient Boosting (Original Data)*

Figure 39: Training Performance Comparison

Figure 40: Validation Performance Comparison

Figure 41: Final Model Selection and Evaluation

Figure 42: Feature Importances

Problem Definition

The United States faces a continuous demand for skilled human resources. To remain competitive, businesses must identify and attract qualified individuals both locally and from abroad. The Immigration and Nationality Act (INA) governs the entry of foreign workers into the US, balancing the protection of domestic workers' wages and working conditions with the need to fill workforce shortages.

The Office of Foreign Labor Certification (OFLC) manages this process by reviewing employer applications for labor certifications. In FY 2016 alone, OFLC processed 775,979 employer applications covering nearly 1.7 million positions, a 9% increase from the previous year. This growth has made the manual review of cases increasingly tedious and time-consuming, creating the need for data-driven, automated support in shortlisting candidates with a higher likelihood of visa approval.

Objective

OFLC has engaged EasyVisa, a data-driven consulting firm, to build a machine learning classification model that can:

- Predict whether a visa application is likely to be certified or denied.
- Identify the key factors (drivers) that influence approval outcomes.
- Provide actionable recommendations to employers for improving their chances of successful certification.

This solution will help streamline the decision-making process, reduce workload for OFLC officers, and enable employers to align their applications with certification requirements more effectively.

Data Description

The data contains the different attributes of the employee and the employer. The detailed data dictionary is given below.

- **case_id**: ID of each visa application
- **continent**: Information of continent the employee
- **education_of_employee**: Information of education of the employee
- **has_job_experience**: Does the employee have any job experience? Y= Yes; N = No
- **requires_job_training**: Does the employee require any job training? Y = Yes; N = No
- **no_of_employees**: Number of employees in the employer's company
- **yr_of_estab**: Year in which the employer's company was established
- **region_of_employment**: Information of foreign worker's intended region of employment in the US.
- **prevailing_wage**: Average wage paid to similarly employed workers in a specific occupation in the area of intended employment. The purpose of the prevailing wage is to ensure that the foreign worker is not underpaid compared to other workers offering the same or similar service in the same area of employment.
- **unit_of_wage**: Unit of prevailing wage. Values include Hourly, Weekly, Monthly, and Yearly.
- **full_time_position**: Is the position of work full-time? Y = Full-Time Position; N = Part-Time Position
- **case_status**: Flag indicating if the Visa was certified or denied

Data Description and Initial Observations

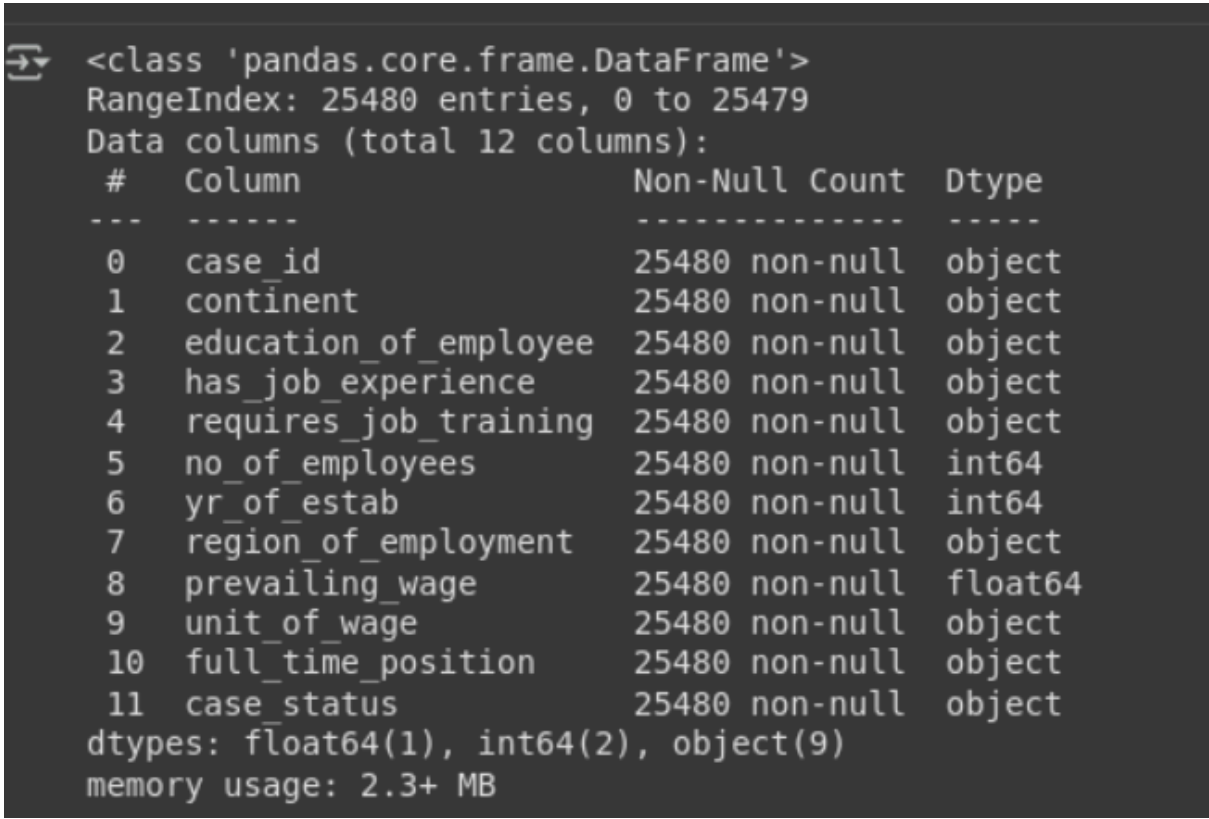


Figure 1: Data Overview

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
case_id	25480	25480	EZYV01	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
continent	25480	6	Asia	16861	NaN	NaN	NaN	NaN	NaN	NaN	NaN
education_of_employee	25480	4	Bachelor's	10234	NaN	NaN	NaN	NaN	NaN	NaN	NaN
has_job_experience	25480	2	Y	14802	NaN	NaN	NaN	NaN	NaN	NaN	NaN
requires_job_training	25480	2	N	22525	NaN	NaN	NaN	NaN	NaN	NaN	NaN
no_of_employees	25480.000	NaN	NaN	NaN	5667.043	22877.929	-26.000	1022.000	2109.000	3504.000	602069.000
yr_of_estab	25480.000	NaN	NaN	NaN	1979.410	42.367	1800.000	1976.000	1997.000	2005.000	2016.000
region_of_employment	25480	5	Northeast	7195	NaN	NaN	NaN	NaN	NaN	NaN	NaN
prevailing_wage	25480.000	NaN	NaN	NaN	74455.815	52815.942	2.137	34015.480	70308.210	107735.513	319210.270
unit_of_wage	25480	4	Year	22962	NaN	NaN	NaN	NaN	NaN	NaN	NaN
full_time_position	25480	2	Y	22773	NaN	NaN	NaN	NaN	NaN	NaN	NaN
case_status	25480	2	Certified	17018	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Figure 2: Statistical Summary

The dataset provided contains **25,480 visa application records** with **12 attributes**, including applicant details, employer characteristics, and job specifications. The objective variable is **case_status**, which indicates whether the visa application was **Certified** or **Denied**.

Target Variable

- **Certified:** 17,018 cases ($\approx 67\%$)
- **Denied:** 8,462 cases ($\approx 33\%$)

This shows a moderate class imbalance, which justifies experimenting with both **oversampling** and **undersampling** strategies in addition to training on the original dataset.

Feature Summary

1. Identifiers

- **case_id:** Unique identifier for each application. It does not add predictive value and has been dropped from modeling.

2. Applicant Characteristics

- **continent:** Six continents represented, with **Asia (16,861 cases)** as the majority, followed by Europe (3,732) and North America (3,292).
- **education_of_employee:** Four levels – Bachelor's (10,234), Master's (9,634), High School (3,420), and Doctorate (2,192). Bachelor's degree is the most common.
- **has_job_experience:** Binary variable (Yes = 14,802; No = 10,678).
- **requires_job_training:** Binary variable (Yes = 2,955; No = 22,525).

3. Employer Characteristics

- **no_of_employees:** Wide range from very small employers to very large (min = 26, max = 602,069).
- **yr_of_estab:** Ranges from 1800 to 2016, covering both newly formed companies and very old institutions.

4. Job Characteristics

- **region_of_employment:** Five regions, with **Northeast (7,195 cases)** and **South (7,017 cases)** leading.

- **prevailing_wage**: Continuous variable, highly skewed, ranging from **2.1 to 319,210**, with mean $\approx 74,456$ and median $\approx 70,308$. This feature will require normalization and possibly log transformation due to extreme values.
- **unit_of_wage**: Four categories – Yearly (22,962), Hourly (2,157), Weekly (272), and Monthly (89).
- **full_time_position**: Binary variable, with **Full-time (22,773)** dominating over Part-time (2,707).

Data Quality Checks

- **Missing Values**: None detected across all variables.
- **Duplicates**: No duplicate records found.
- **Data Types**:
 - Numeric: **no_of_employees**, **yr_of_estab**, **prevailing_wage**.
 - Categorical: **continent**, **education_of_employee**, **has_job_experience**, **requires_job_training**, **region_of_employment**, **unit_of_wage**, **full_time_position**.
 - Target: **case_status** (binary categorical).

Key Observations from Initial Audit

- The dataset is **clean**: no missing values, no duplicates, consistent data types.
- **prevailing_wage** shows a heavy-tailed distribution, requiring **scaling/log transformation**.
- Class imbalance exists (67:33), so resampling strategies will likely improve recall for the minority class.
- Certain categorical variables like **continent**, **education_of_employee**, and **region_of_employment** show natural groupings that may strongly influence visa approval likelihood.

Exploratory Data Analysis

Univariate Analysis

- Histograms & box-plots for numeric columns.
- For categorical columns we use bar (count) plots or a frequency table.

1.continent

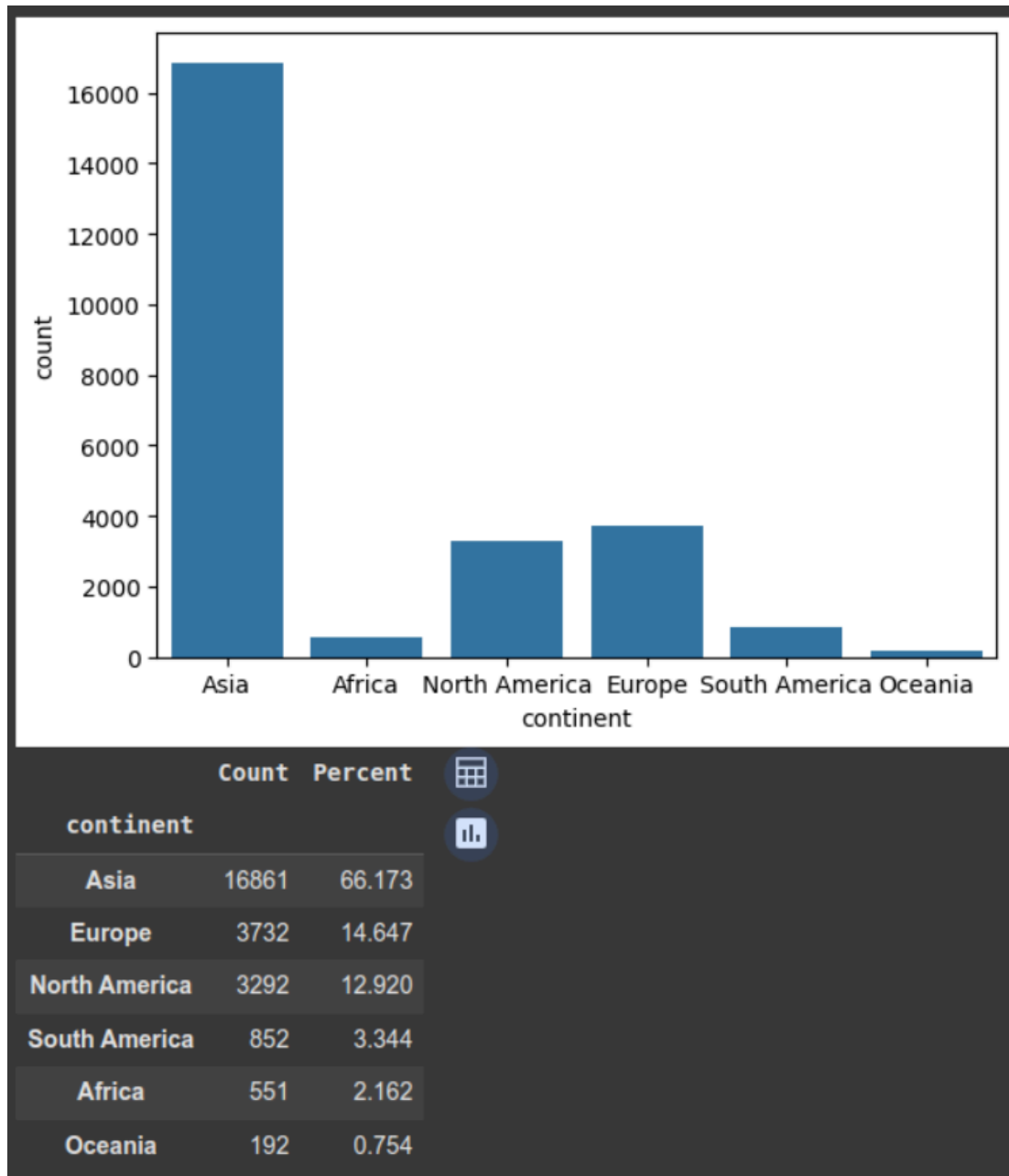


Fig.3: Barplot and frequency table of continent

The variable **continent** represents the applicant's continent of origin. The distribution is highly skewed, with **Asia dominating the dataset**.

- **Asia** accounts for the majority of applications, with **16,861 cases (≈66%)**.
- **Europe** is the second-largest contributor, with **3,732 cases (≈15%)**.
- **North America** follows closely with **3,292 cases (≈13%)**.
- Other regions such as **South America (≈3.3%)**, **Africa (≈2.2%)**, and **Oceania (≈0.7%)** make up only a small fraction of the dataset.

Key Observation

- The dataset is **dominated by Asian applicants**, reflecting the real-world trend of high demand for US employment visas from Asian countries (particularly India and China).
- Regions like **Oceania and Africa are underrepresented**, meaning their influence on model learning will be minimal compared to Asia, Europe, and North America.

2. education_of_employee

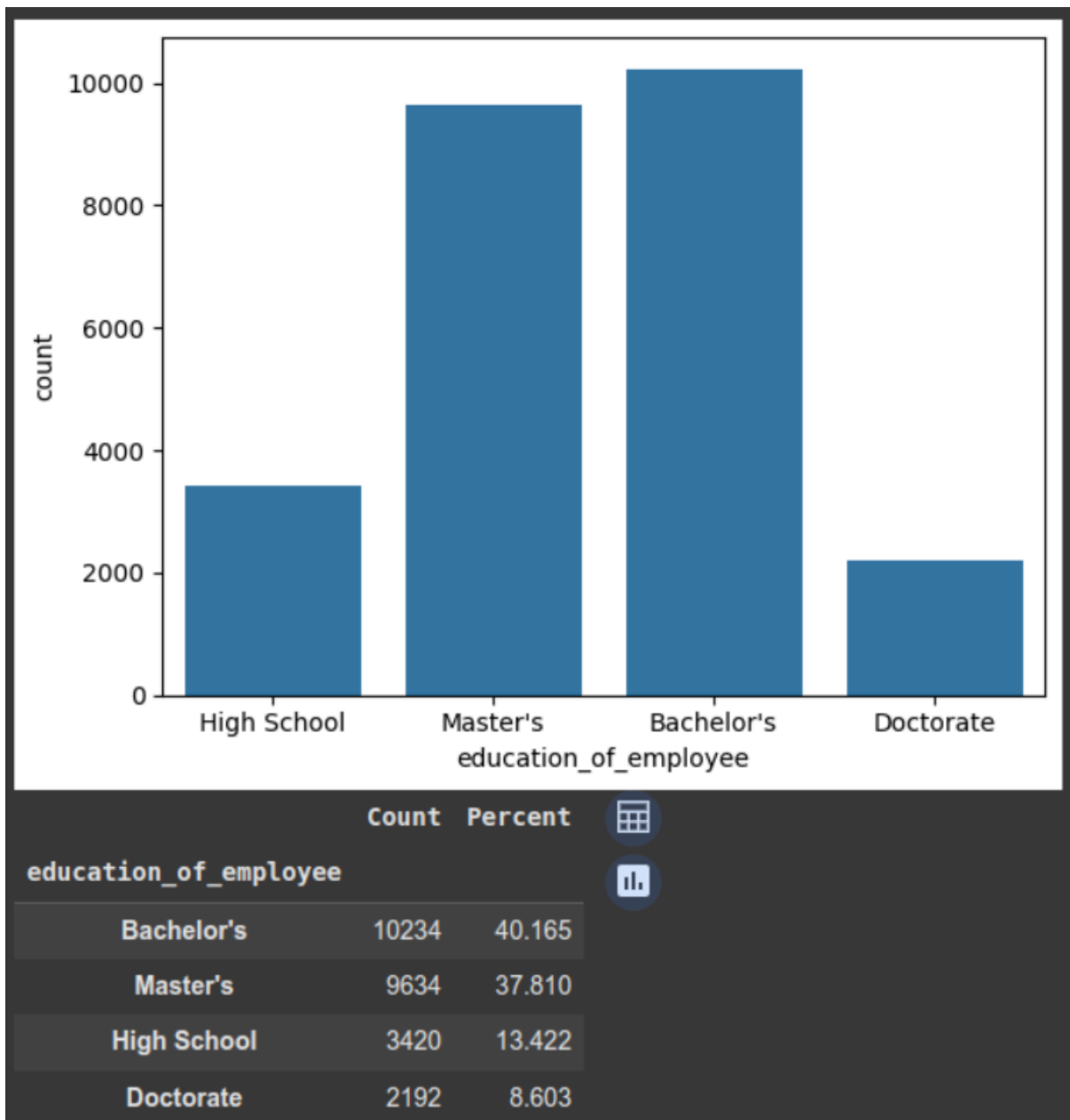


Fig.4: Barplot and frequency table of education_of_employee

The variable **education_of_employee** represents the highest educational qualification of the applicants. The distribution shows that most applicants hold either a **Bachelor's** or a **Master's** degree.

- **Bachelor's**: 10,234 applicants (**40.2%**)
- **Master's**: 9,634 applicants (**37.8%**)
- **High School**: 3,420 applicants (**13.4%**)

- **Doctorate:** 2,192 applicants (8.6%)

Key Observation

- The majority of applicants are **highly educated**, with nearly **78% holding at least a Bachelor's or Master's degree**.
- A relatively smaller portion of applicants have **Doctorates (8.6%)**, while **High School graduates (13.4%)** make up the least among non-advanced qualifications.
- This suggests that visa applications are predominantly from **skilled and highly qualified workers**, which aligns with the goal of the US immigration system to attract specialized talent.

3. has_job_experience

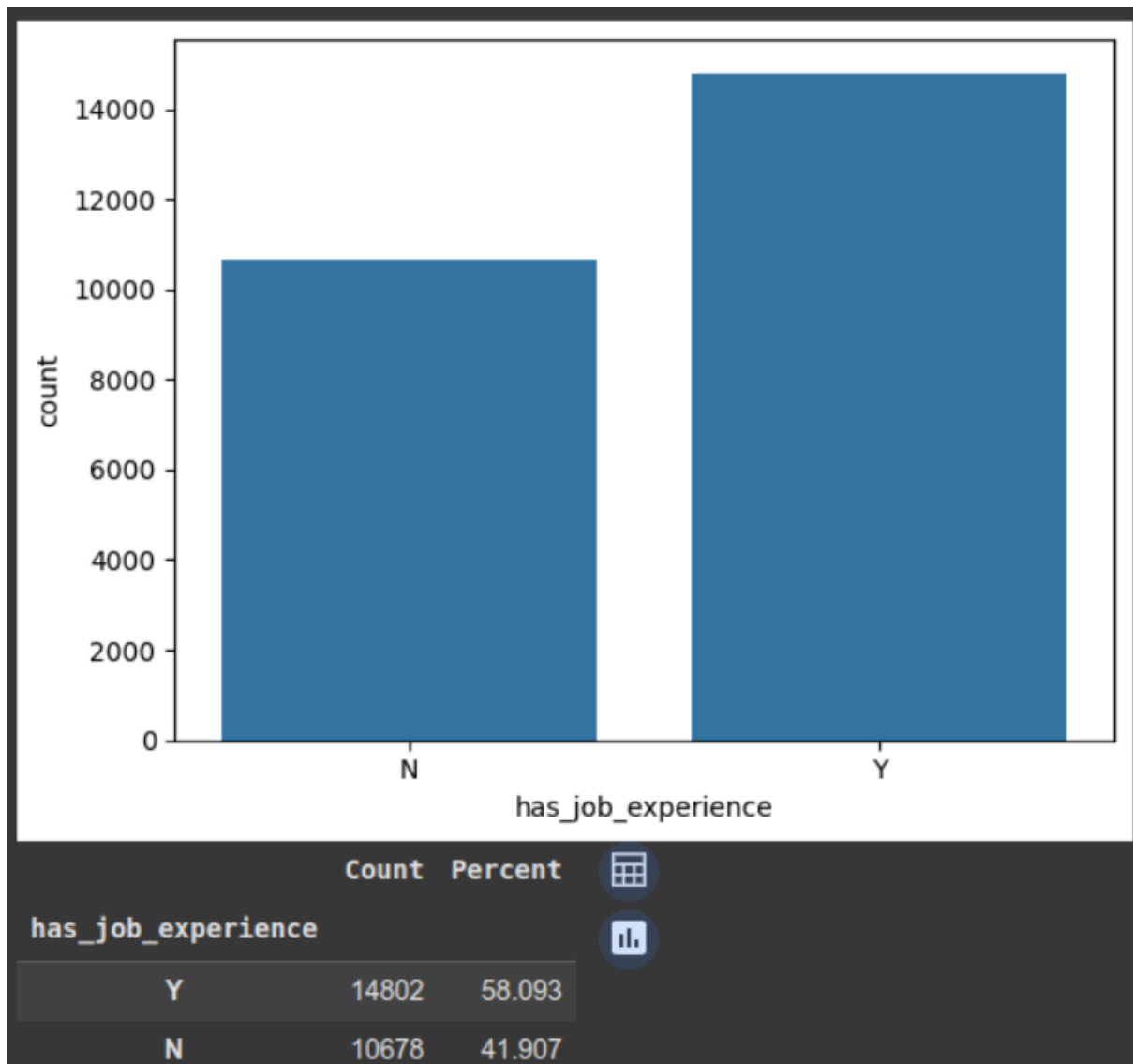


Fig.5: Barplot and frequency table of has_job_experience

A majority of applicants reported having prior job experience.

- **Yes (Y):** 14,802 applicants (**58.1%**)
- **No (N):** 10,678 applicants (**41.9%**)

Key Observation

- The dataset shows a strong representation of **experienced workers**, which aligns with the US labor certification system's preference for skilled employees.

- Still, a significant portion ($\approx 42\%$) of applicants do not have job experience, suggesting that the dataset includes both **entry-level candidates** and **professionals**.

4. requires_job_training

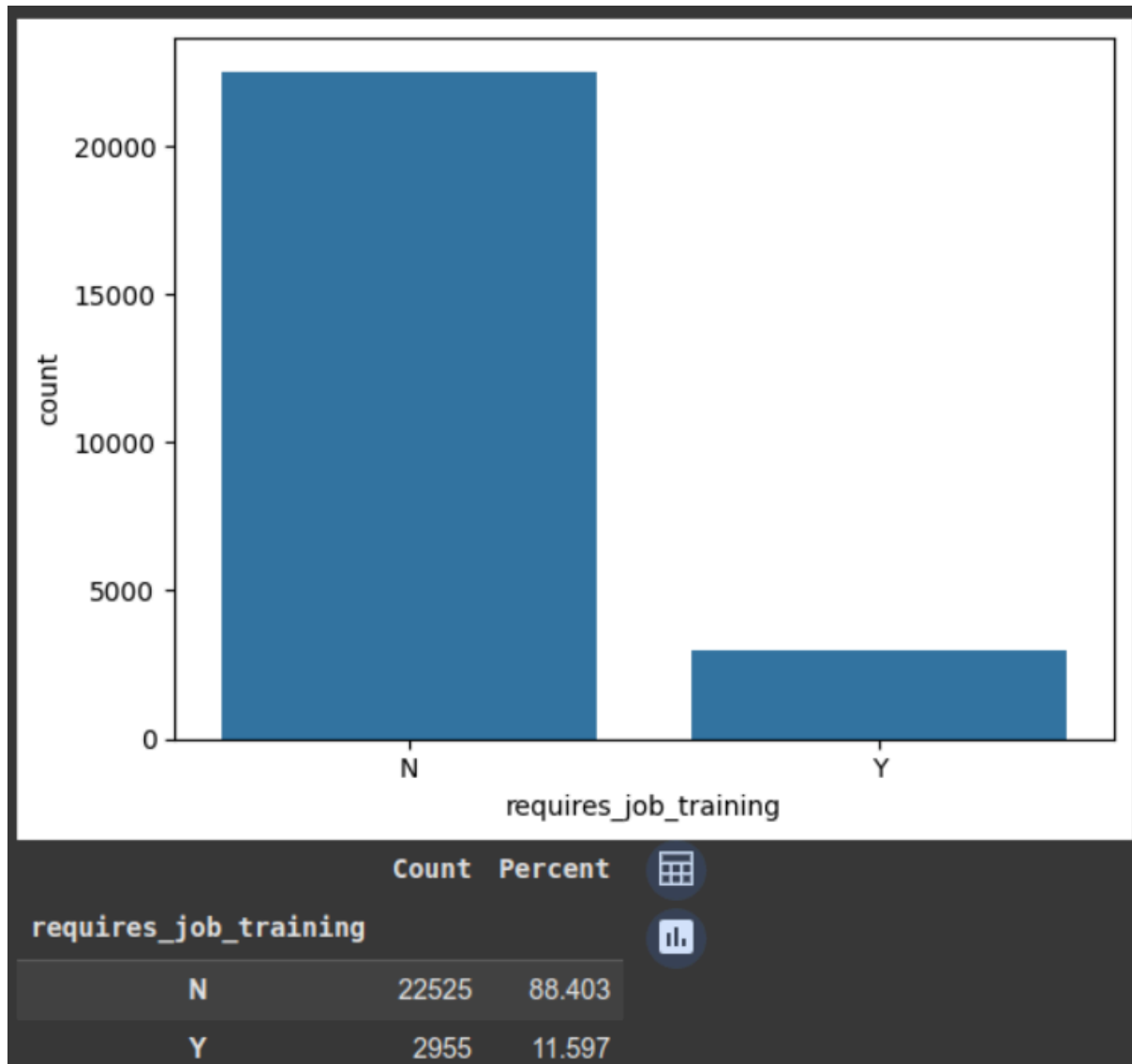


Fig.6: Barplot and frequency table of requires_job_training

Most applicants do not require additional job training.

- **No (N):** 22,525 applicants (**88.4%**)
- **Yes (Y):** 2,955 applicants (**11.6%**)

Key Observation

- The dataset is heavily skewed toward applicants who are **already job-ready**, which reduces the burden on employers for extra training.
- Only a small minority require training, indicating that **specialized or niche roles** may account for this group. This variable could be an important differentiator in visa approval chances.

5. no_of_employees

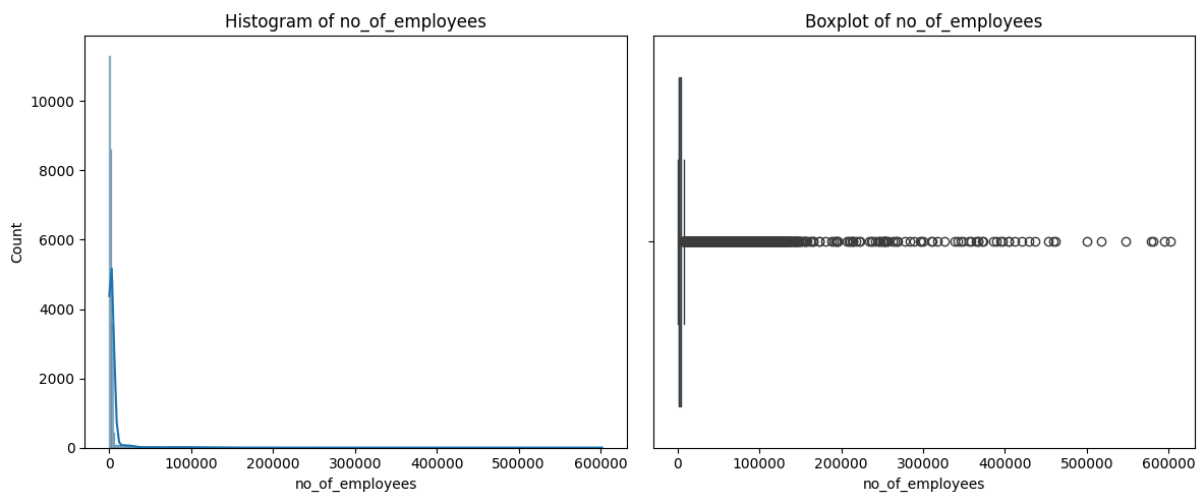


Fig.7: Histogram and Boxplot of no_of_employees

The variable **no_of_employees** represents the workforce size of the sponsoring company. The distribution is **heavily right-skewed**, with a few extremely large firms creating long tails in the data.

- The majority of employers have relatively **small to medium-sized workforces**, concentrated below 20,000 employees.
- A handful of companies report **employee counts above 100,000**, with the maximum reaching nearly **600,000 employees**.

Key Observation

- The skewed distribution indicates the presence of **significant outliers** (very large corporations).
- Most cases cluster around smaller firms, but the extreme values will likely need **log transformation** or **winsorization** during preprocessing to stabilize their influence in the model.

- Company size could still be a meaningful predictor, since larger, established firms may face fewer hurdles in visa certifications compared to small or newly established companies.

6. yr_of_estab

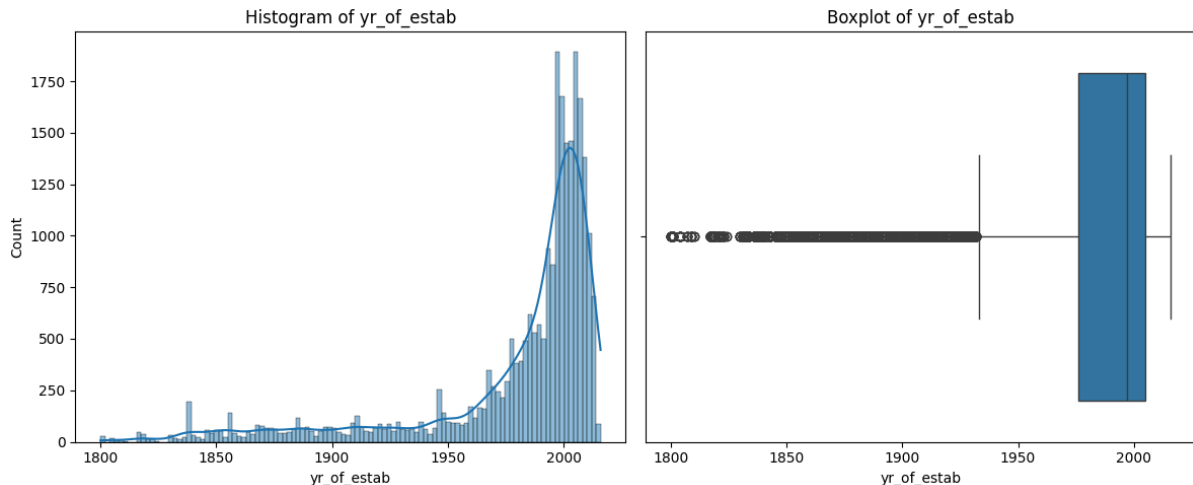


Fig.8: Histogram and Boxplot of yr_of_estab

The variable **yr_of_estab** indicates the year the sponsoring company was established. The distribution shows a long history of company formations, stretching back to the 1800s.

- Most companies in the dataset were founded in the **second half of the 20th century (1950–2000)**, with a peak close to the **1990s and early 2000s**.
- A few companies trace their origins back to the **early 1800s**, which appear as extreme outliers in the data.
The boxplot confirms a strong **right-skewness**, with older companies being rare compared to the majority that were established in recent decades.

Key Observation

- The raw **yr_of_estab** is less interpretable than transforming it into **employer_age (2016 – yr_of_estab)**, which directly measures how long the company had been in operation at the time of application.

- Extremely old values (over 150–200 years) should be flagged as potential **outliers**, as they may distort modeling without appropriate scaling or capping.
- The variable is likely meaningful, since **longer-established companies may have more credibility and higher visa approval rates**.

7. employer_age

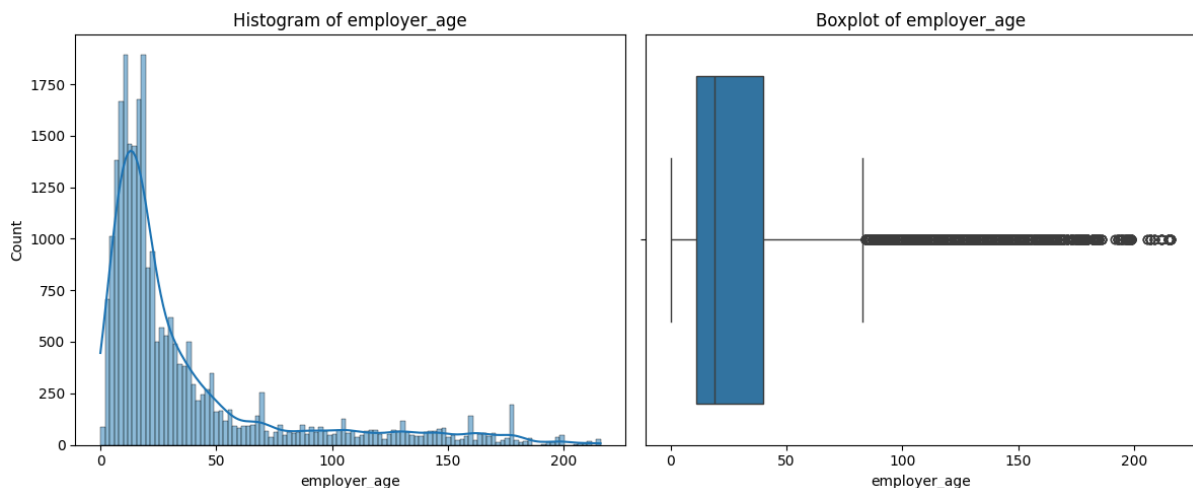


Fig.9: Histogram and Boxplot of employer_age

The dataset originally contained the variable **yr_of_estab**, which represented the year when the sponsoring employer was established. While this variable is informative, in its raw form it is not directly interpretable in a business context. To make it more meaningful, we transformed it into a new variable:

$$\text{employer_age} = 2016 - \text{yr_of_estab}$$

Here, 2016 is used because the data corresponds to **Fiscal Year 2016**. This conversion allows us to measure the age of the company — how long the employer had been operating at the time of the visa application.

After creating the new variable, we **dropped** **yr_of_estab** to avoid redundancy and potential multicollinearity during model building.

Why we did this

- **Interpretability:** Business stakeholders can more easily understand “a company is 20 years old” rather than “founded in 1996.”
- **Business relevance:** Employer age is a strong indicator of company stability and credibility. More established employers may have a higher probability of visa approvals due to stronger compliance records and financial security.
- **Statistical usefulness:** Unlike raw establishment years, employer age is a continuous variable on a meaningful scale, making it more suitable for both descriptive analysis and predictive modeling.

What we observed

- The majority of employers fall between **10 and 50 years old**, peaking in the range of **15–25 years**.
- A significant number of employers are more than **100 years old**, with extreme cases exceeding **200 years**.
- These very old values appear as **outliers** in the distribution. While they are not invalid (since some large multinational corporations have been around for over a century), they may distort the model and require **outlier capping or log transformation** during preprocessing.

Key Takeaway

Transforming `yr_of_estab` into `employer_age` makes the feature both **business-relevant** and **statistically sound**. It provides a direct measure of organizational maturity, which is likely to influence visa approval outcomes.

8. region_of_employment

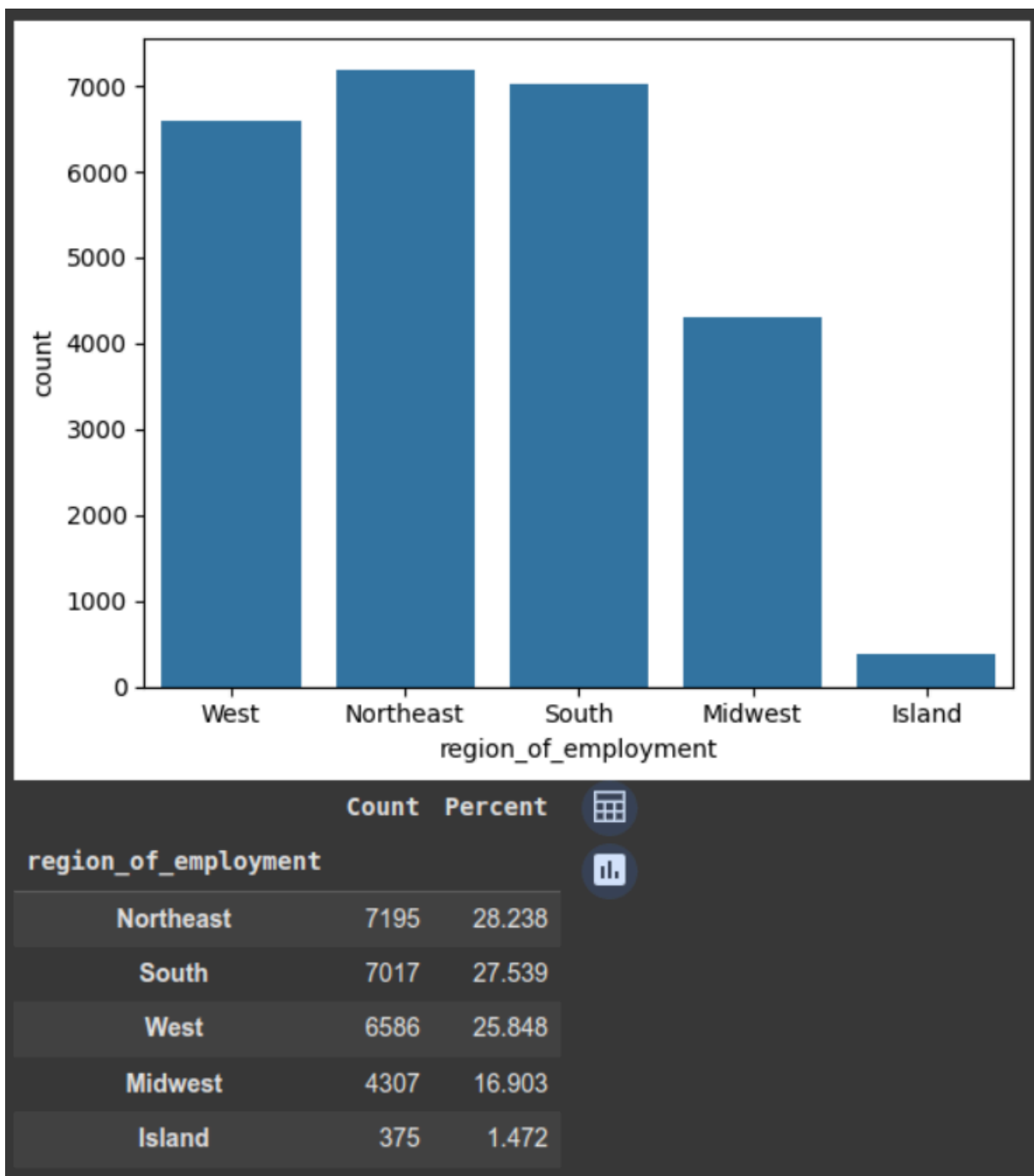


Fig.10: Barplot and frequency table of `region_of_employment`

The variable `region_of_employment` indicates the U.S. region where the foreign worker is expected to be employed. The distribution across regions is fairly balanced, with some mild variation.

- **Northeast:** 7,195 applications (**28.2%**)
- **South:** 7,017 applications (**27.5%**)
- **West:** 6,586 applications (**25.8%**)
- **Midwest:** 4,307 applications (**16.9%**)
- **Island (e.g., Hawaii, Puerto Rico):** 375 applications (**1.5%**)

Key Observation

- The **Northeast, South, and West** regions account for over **80% of all applications**, reflecting their higher concentration of economic and technological activity.
- The **Midwest** shows moderate representation, while the **Island region** contributes very few cases.
- This variable could capture regional labor market dynamics—visa approvals may vary depending on workforce shortages and wage norms in different U.S. regions.

9. prevailing_wage

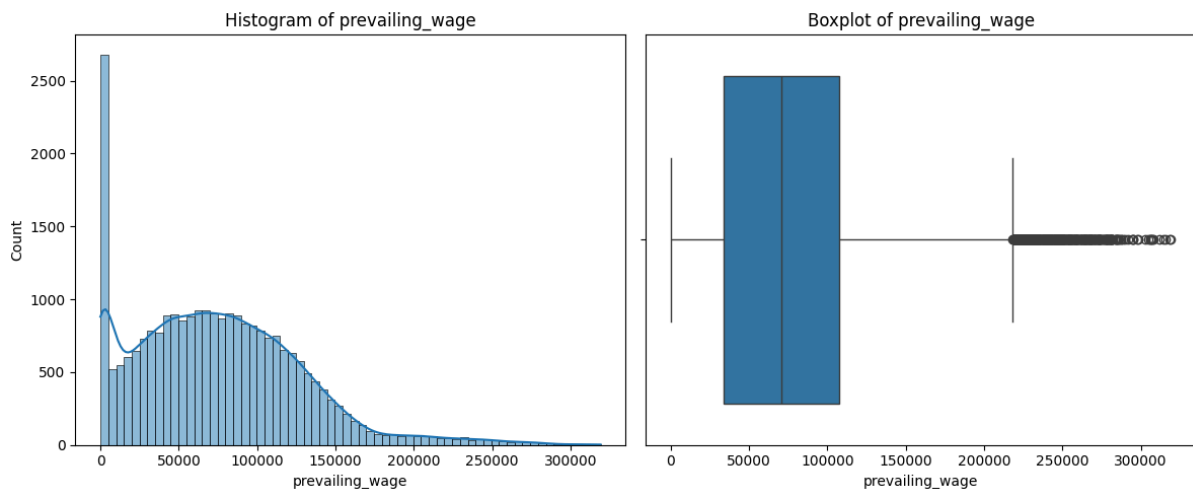


Fig.11: Histogram and Boxplot of prevailing_wage

The variable **prevailing_wage** represents the average wage paid to similarly employed workers in the same occupation and region. It ensures that foreign workers are not paid less than U.S. workers performing comparable jobs.

From the histogram and boxplot, the distribution of prevailing wages is **positively skewed**, with most values concentrated at the lower end and a long tail extending toward very high salaries.

- The majority of wages fall roughly between **\$30,000 and \$120,000**, with a noticeable peak in the lower-middle range.
- A smaller number of cases exceed **\$200,000**, reaching a maximum of around **\$300,000**, which appear as outliers.
- The median wage appears to be near **\$70,000–\$80,000**, consistent with mid-level professional salaries.

Key Observation

- The skewed distribution indicates that while most applications correspond to **mid-level technical or professional roles**, there are a few very high-paying positions—likely **senior or specialized technical jobs**.
- These high-end values may need **outlier treatment or scaling (e.g., log transformation)** before modeling.
- From a business standpoint, **prevailing wage is likely a strong predictor of visa approval**, as higher wages often signal specialized skills and help employers demonstrate that hiring foreign workers will not undercut local wages.

10. unit_of_wage

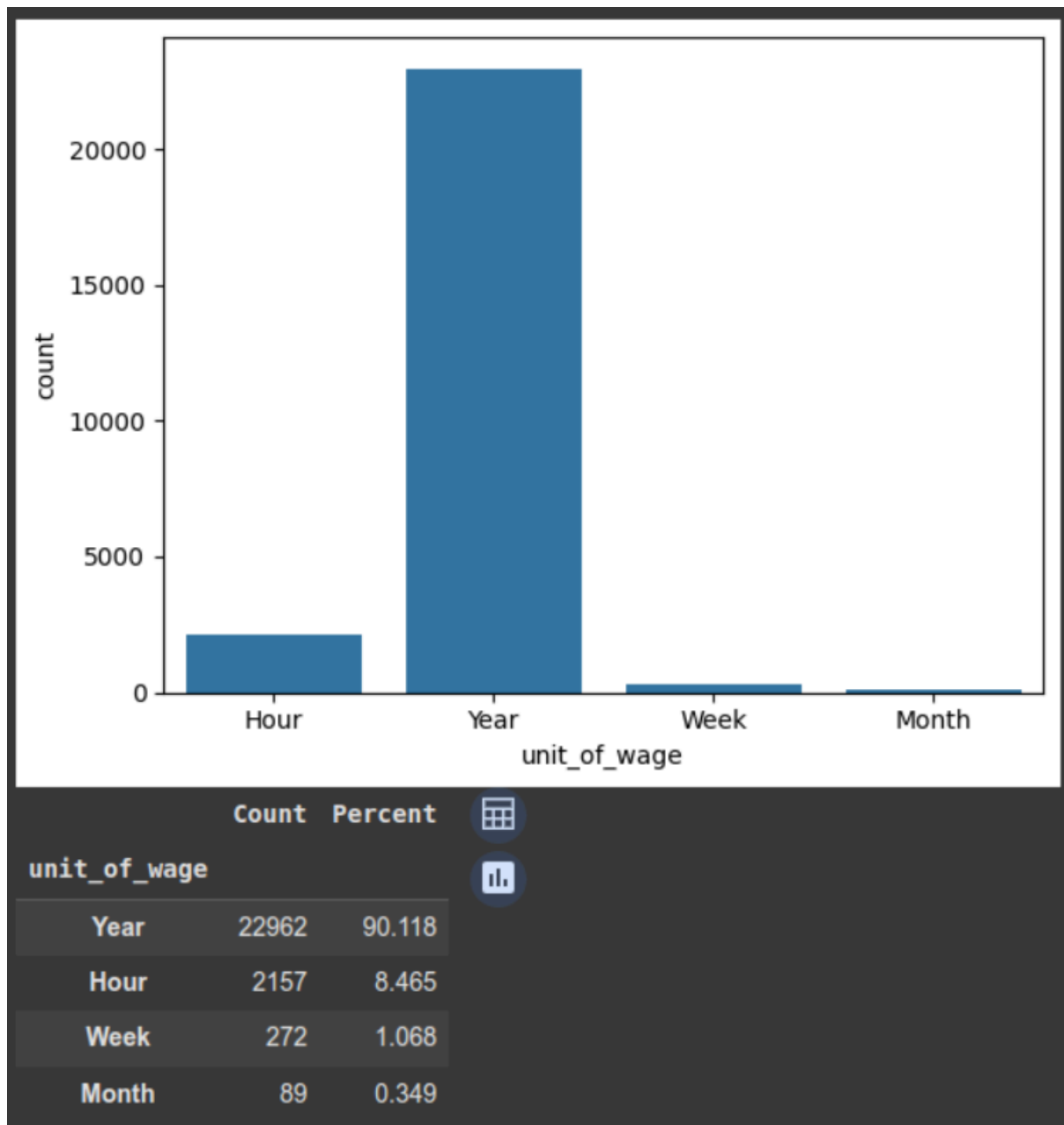


Fig.12: Barplot and frequency table of region_of_employment

The variable `unit_of_wage` specifies the pay frequency used to report the prevailing wage — whether the salary is expressed hourly, weekly, monthly, or yearly. The distribution is **heavily dominated by yearly wages**.

- **Year:** 22,962 cases (90.1%)
- **Hour:** 2,157 cases (8.5%)
- **Week:** 272 cases (1.1%)

- **Month:** 89 cases (**0.3%**)

Key Observation

- The overwhelming majority of wage records are reported on a **yearly basis**, which aligns with the nature of most **full-time professional and technical positions** that the visa program caters to.
- Hourly wages represent a smaller subset, likely corresponding to **temporary or part-time roles**, whereas weekly and monthly wage reports are rare.
- Since the data is highly imbalanced across categories, the **unit_of_wage** variable may be simplified or encoded carefully during preprocessing to avoid undue model bias.
- This feature can be an indirect indicator of **employment type and job stability**, which might influence the likelihood of visa certification.

11. full_time_position

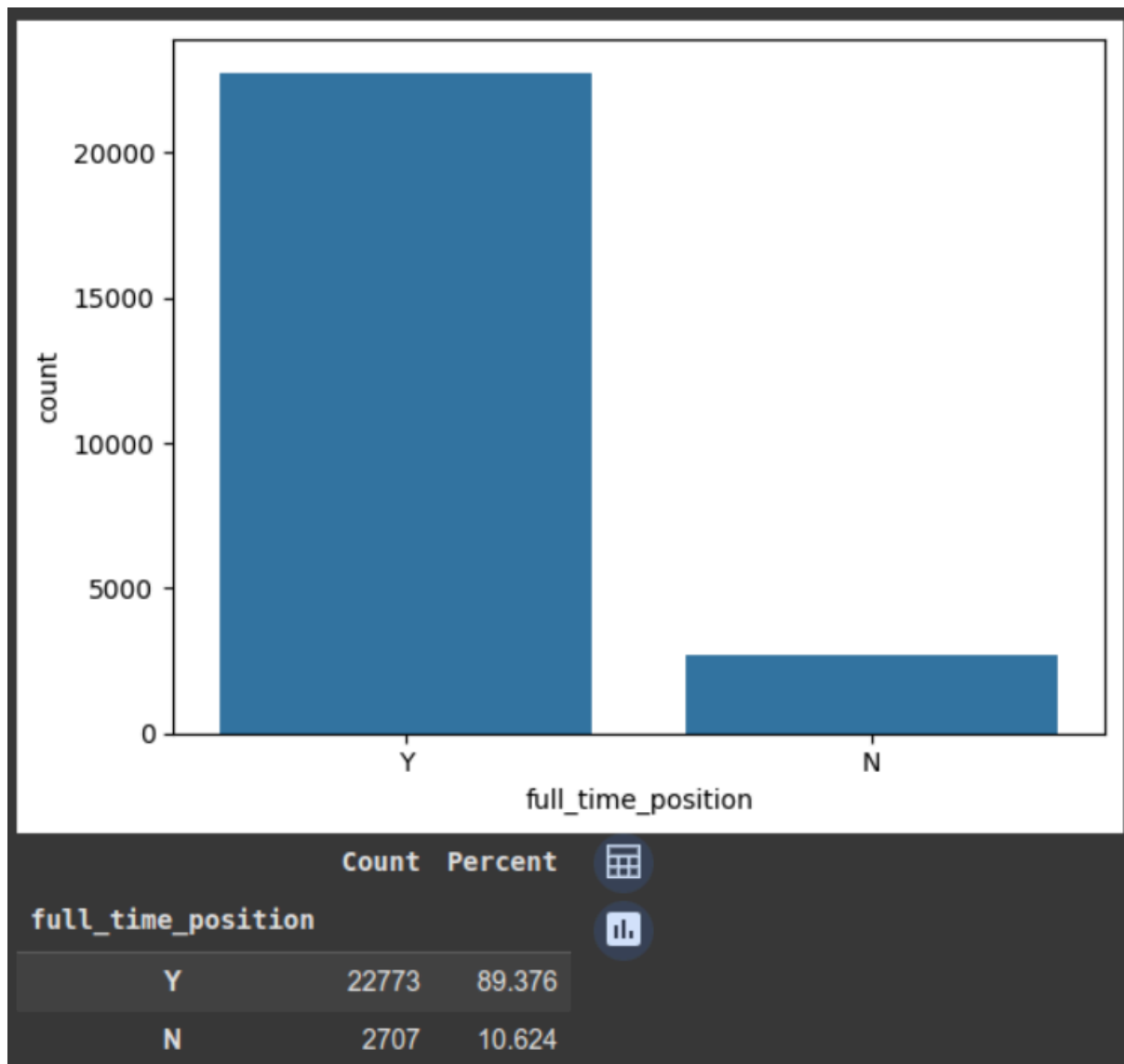


Fig.13: Barplot and frequency table of `full_time_position`

The variable `full_time_position` indicates whether the offered employment is a full-time or part-time role. The distribution shows a clear dominance of full-time positions among visa applications.

- **Yes (Y):** 22,773 cases (**89.4%**)
- **No (N):** 2,707 cases (**10.6%**)

Key Observation

- The overwhelming majority of visa applications are for **full-time positions**, highlighting that U.S. employers primarily sponsor

foreign workers for **permanent or long-term roles** rather than part-time work.

- Part-time positions form only a small fraction of the dataset, suggesting limited visa demand in that segment.
- From a business perspective, full-time roles are generally **more likely to be certified**, since they align with the program's intent to fill essential and ongoing labor shortages rather than temporary or flexible work arrangements.

12. case_status

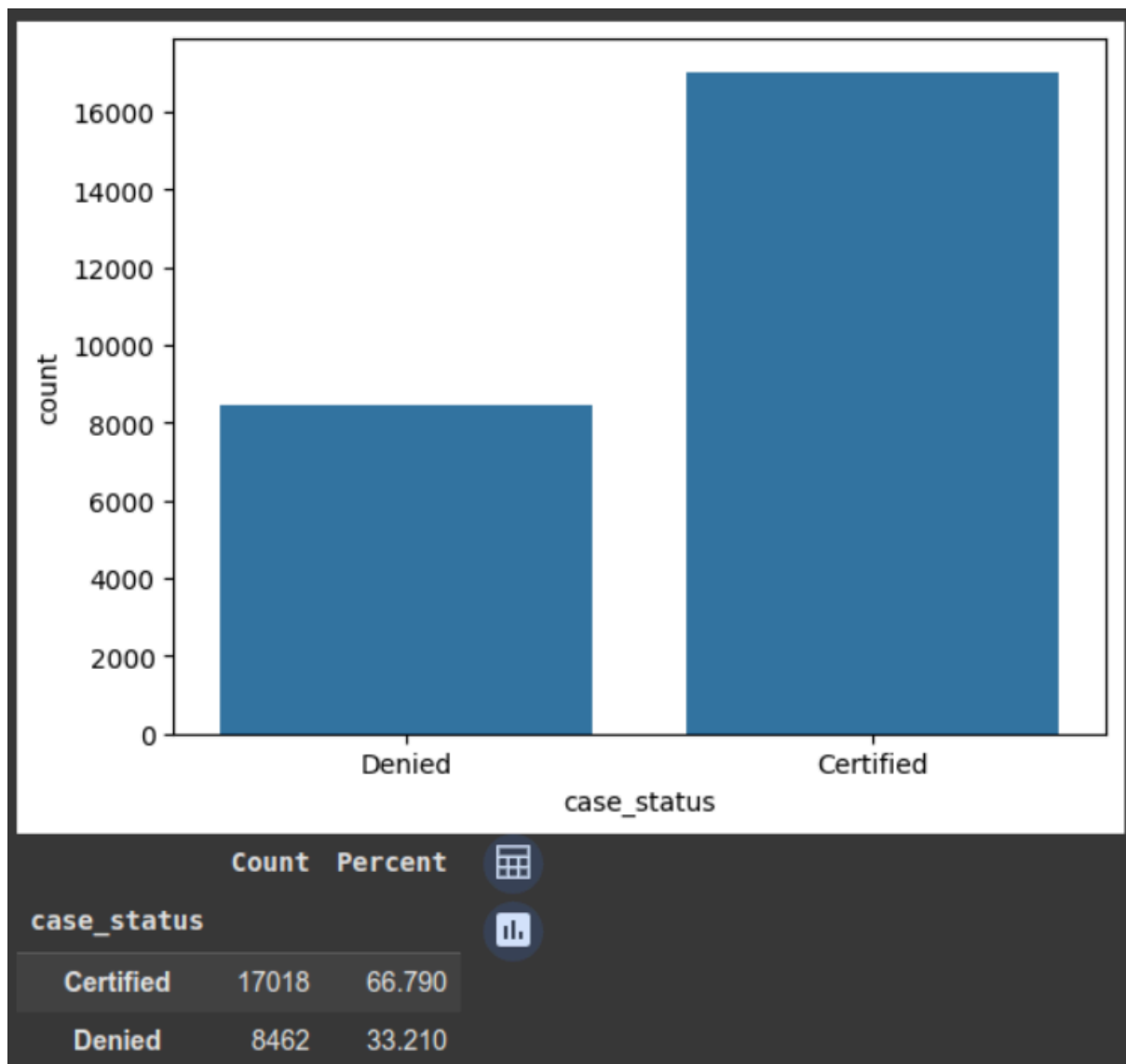


Fig.14: Barplot and frequency table of case_status

The variable **case_status** represents the outcome of each visa application — whether it was **certified (approved)** or **denied**.

- **Certified:** 17,018 applications (**66.8%**)
- **Denied:** 8,462 applications (**33.2%**)

Key Observation

- Approximately **two-thirds of all applications were certified**, while about **one-third were denied**, indicating a **moderate class imbalance** in the target variable.
- This suggests that while most applications succeed, a substantial portion still fail, making it worthwhile to identify the key differentiators between approved and rejected cases.
- From a modeling standpoint, the imbalance is **not severe but notable**, and **resampling techniques (oversampling or undersampling)** may be useful during training to ensure balanced model learning.
- In business terms, understanding the factors driving certification can help both employers and policymakers **improve approval rates by aligning applications with OFLC's standards**.

Bivariate Analysis

Correlation among Numerical Variables

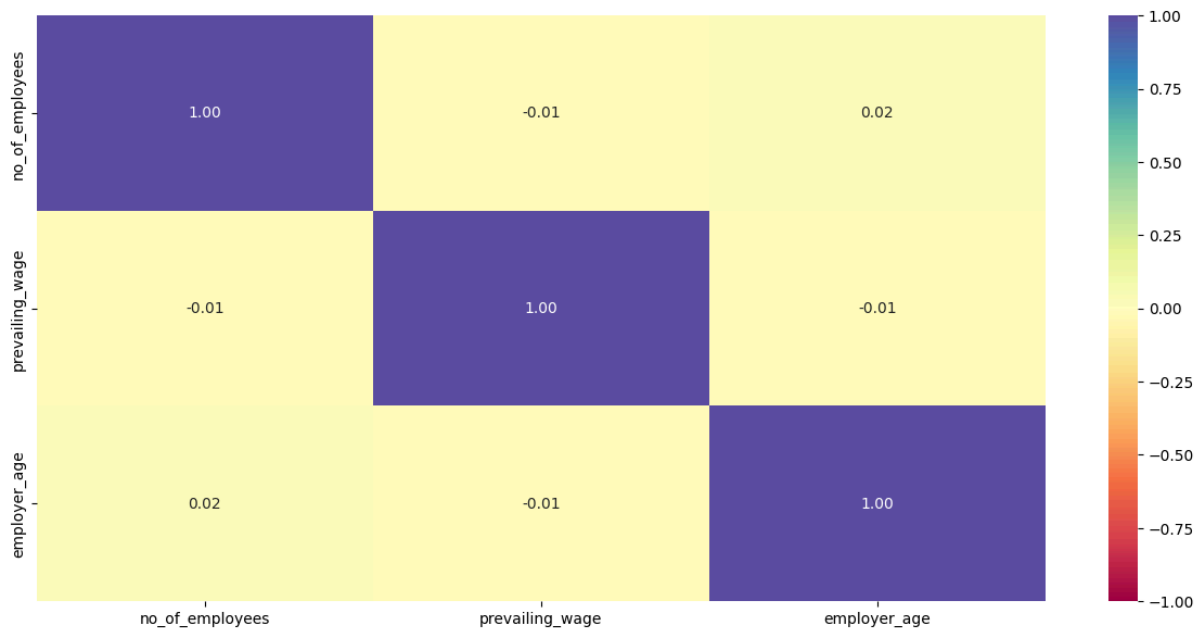


Fig.15: Heat map for numerical variables

The heatmap above illustrates the pairwise correlations between the three numerical variables — **no_of_employees**, **prevailing_wage**, and **employer_age**.

Observations

- All correlation coefficients are **very close to zero** (ranging between -0.01 and 0.02), suggesting **no significant linear relationships** among these numerical features.
- This means that company size, wage level, and employer age are **independent of each other** — for instance, large firms don't necessarily pay higher wages or have longer operational histories.
- The lack of correlation also implies **low risk of multicollinearity**, which is beneficial for model stability and interpretability during the machine learning phase.

Key Insight

Since these variables do not strongly influence one another, each may independently contribute predictive power to the visa certification outcome. Their weak correlation ensures that **no redundant numerical features** will distort model weights or inflate variance during training.

Pair Plot Analysis – Numerical Variables

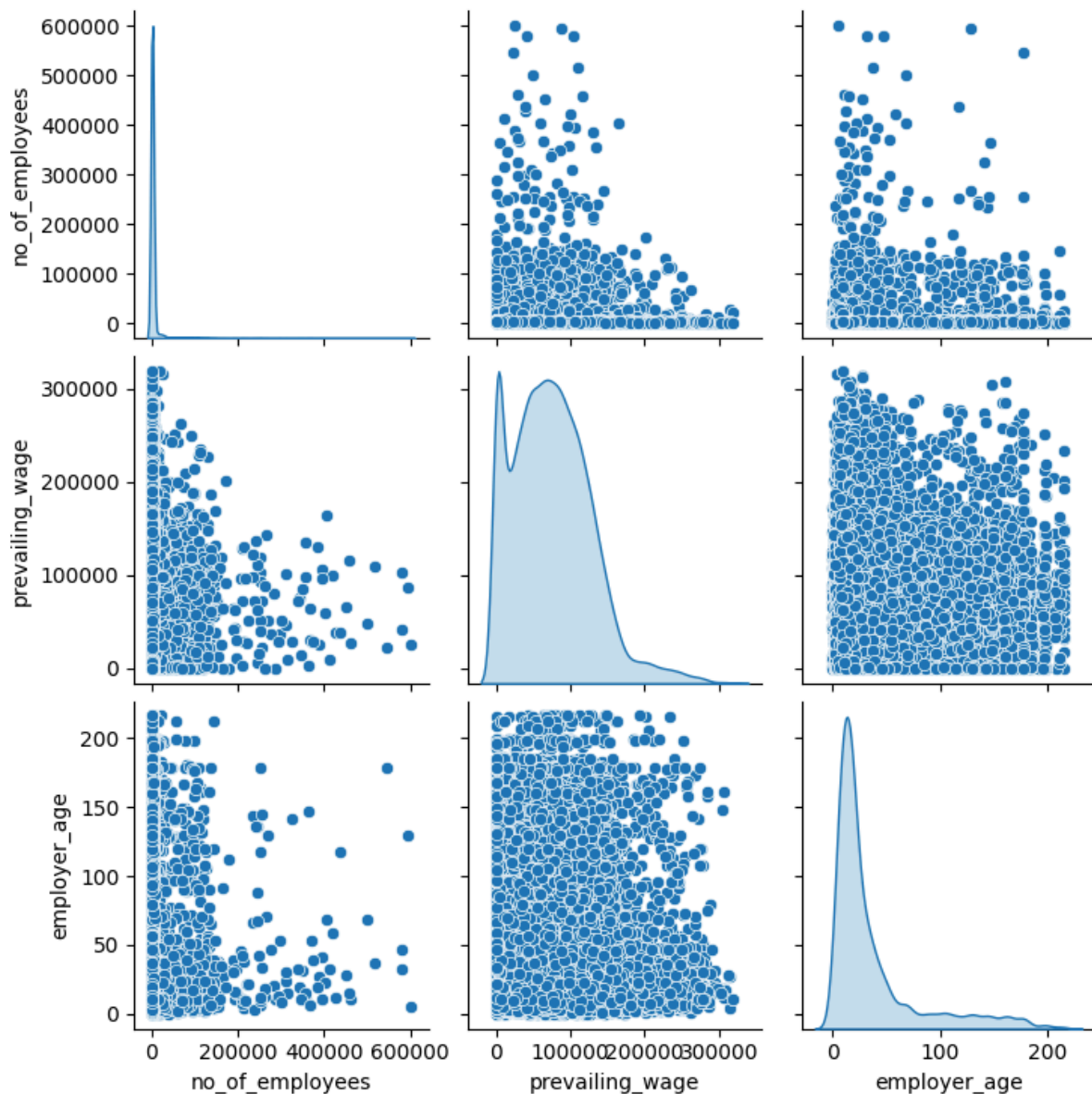


Fig.16: Pairplot for numerical variables

The pair plot visualizes the relationships among the three numerical variables — **no_of_employees**, **prevailing_wage**, and **employer_age** — while also showing their individual distributions along the diagonal.

Observations

- **No clear linear or non-linear relationships** are visible among the numerical features. The scatterplots appear randomly distributed, consistent with the earlier correlation matrix findings.
- The **distribution of no_of_employees** is highly right-skewed, with most firms having relatively small workforces and a few massive outliers employing tens or hundreds of thousands.
- **prevailing_wage** exhibits a somewhat bimodal distribution, possibly reflecting two major clusters of job roles—**mid-level technical jobs** and **high-skill specialized positions**.
- **employer_age** also shows a long right tail, indicating that while most companies are relatively young, a few have been operating for over a century.

Insights

- The lack of any visible pattern or relationship among these features reinforces that **multicollinearity is not a concern**.
- Each numerical variable appears to provide **independent predictive value**, which is advantageous for model interpretability.
- Given the visible skewness, applying **log transformations or outlier capping** may help stabilize variance and improve model performance during training.

Prevailing Wage by Case Status

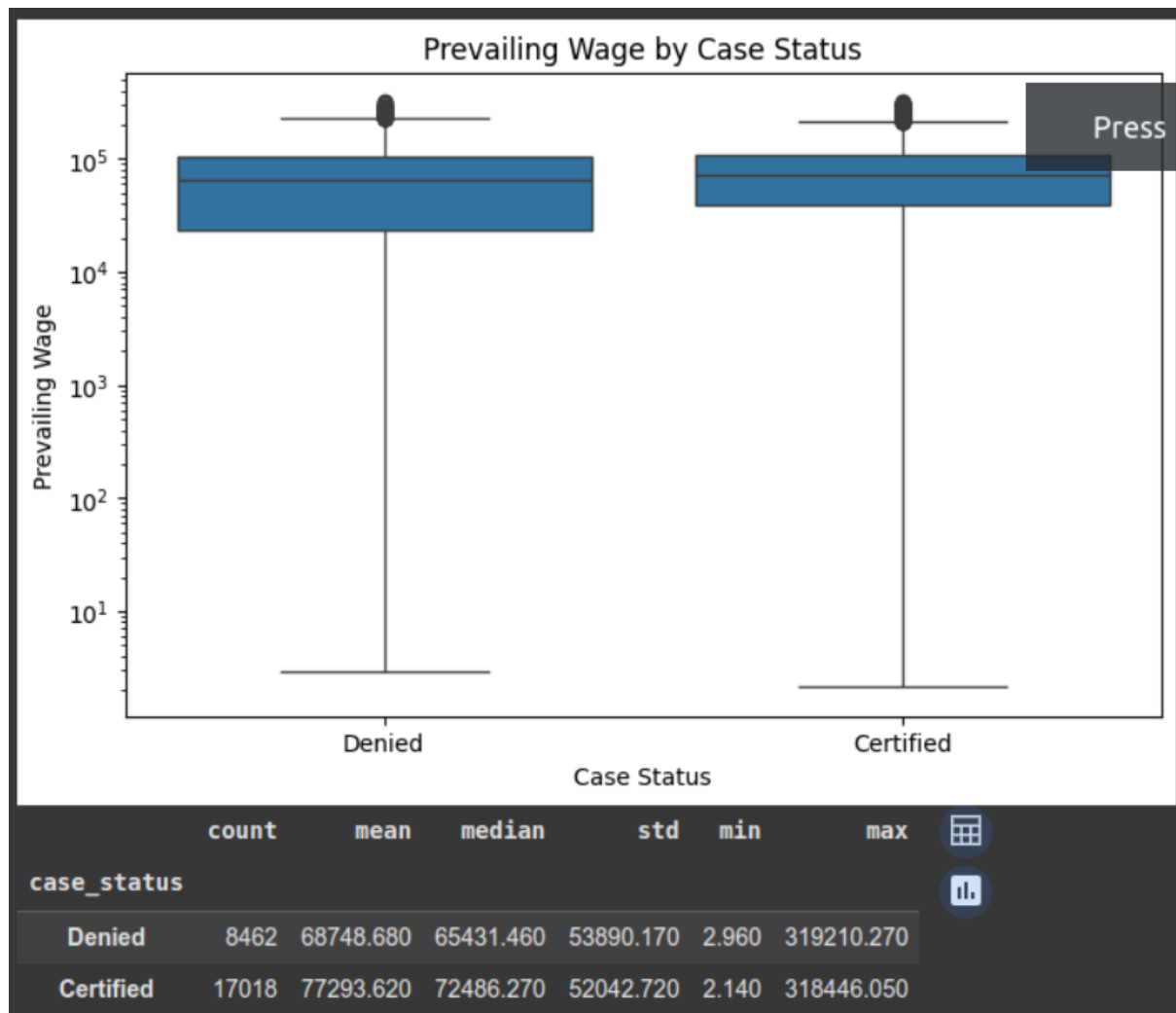


Fig.17: Prevailing Wage by Case Status

The variable **prevailing_wage** represents the average wage paid to similarly employed workers in a specific occupation and location within the U.S. It serves as an important determinant in visa certification, as it ensures that hiring foreign workers does not undercut local wage standards.

Observations

- The **average prevailing wage** for **certified applications** is approximately **\$77,294**, while for **denied cases** it is around **\$68,749**.
- The **median wage** follows the same trend, with **certified cases** showing a higher median (**\$72,486**) than **denied cases** (**\$65,431**).

- The distribution of wages is **right-skewed**, with a long tail of very high-paying positions—these appear as outliers on the boxplot.

Key Observations

- Applicants offered **higher prevailing wages** are **more likely to receive visa certification**, indicating that competitive compensation plays a positive role in approval decisions.
- This likely reflects OFLC's emphasis on ensuring fair pay relative to local wage standards, minimizing the risk of **wage suppression or exploitation** of foreign workers.
- However, the wide spread and outliers suggest that **wage alone is not a definitive factor**—it likely interacts with variables such as **education, experience, and job region** to influence outcomes.

Insight

Higher prevailing wages tend to correlate with visa certification, underscoring that **economic justification and fair pay** remain crucial determinants of successful visa approvals.

Employer Age by Case Status

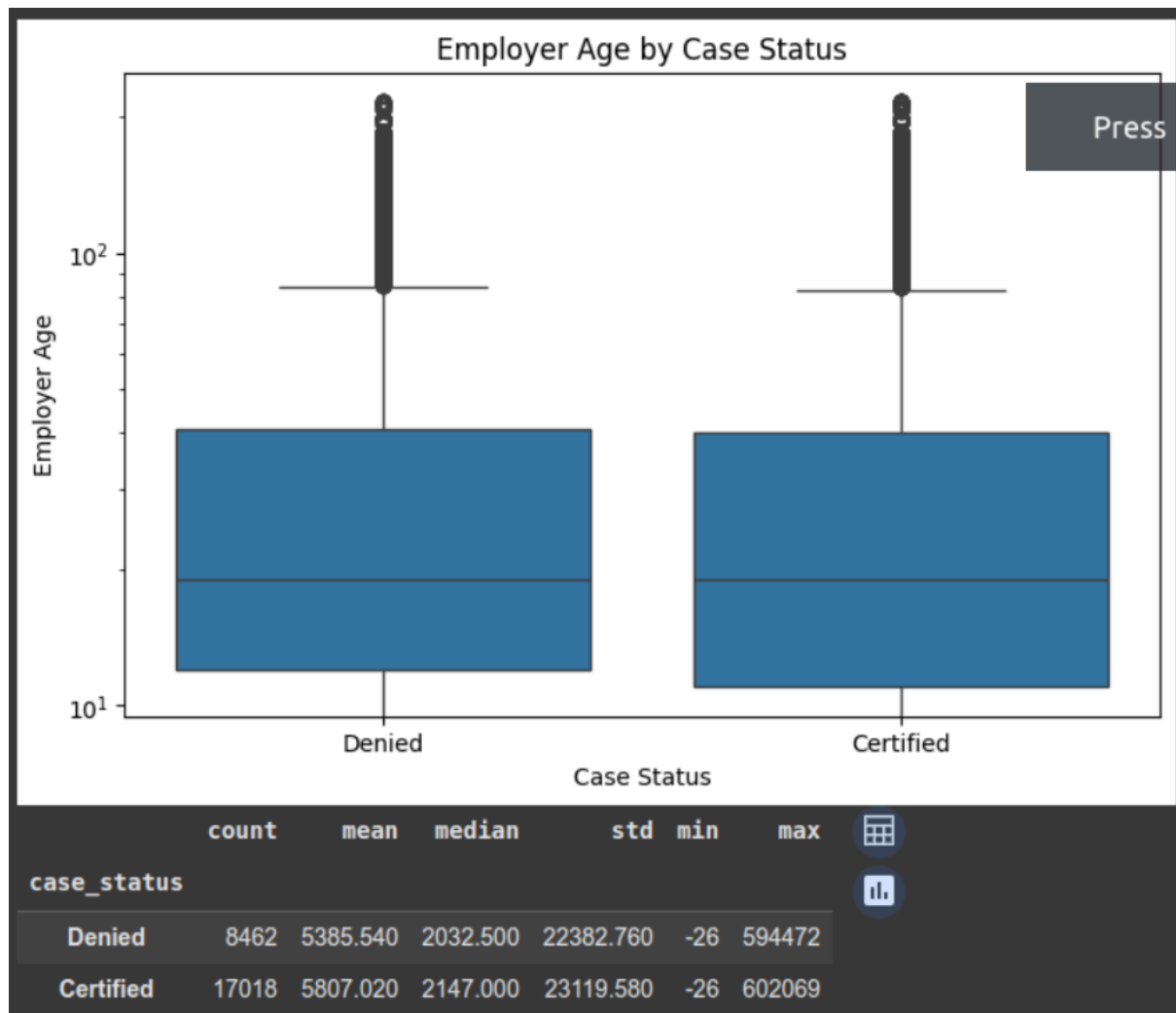


Fig.18: Employer Age by Case Status

The **employer_age** variable represents how long the sponsoring company has been established in years (calculated as $2016 - \text{yr_of_estab}$). This serves as a proxy for organizational stability, credibility, and longevity—factors that might influence visa certification decisions.

Observations

- The **average employer age** is nearly identical for both **certified** (**~29 years**) and **denied** (**~28 years**) applications.

- The **median values** show minimal variation, indicating a consistent distribution across both outcome categories.
- The data is **right-skewed**, with most companies under 50 years old but a few long-established organizations exceeding **100 years**.
- Outliers are expected (representing older corporations) but do not indicate data quality issues.

Key Observations

- The close similarity between certified and denied distributions suggests that **employer age alone has little direct impact** on visa approval.
- While older firms may intuitively seem more credible, the results imply that **current compliance, wage level, and job type** are more influential determinants.
- Nonetheless, **employer_age** may still hold **marginal predictive power** when combined with other employer characteristics such as **no_of_employees** or **region_of_employment**.

Insight

Employer longevity has **minimal standalone effect** on visa certification outcomes. OFLC's decision process appears to prioritize **economic justification and regulatory compliance** over company age, focusing more on the employer's **current standing** than historical legacy.

Number of Employees by Case Status

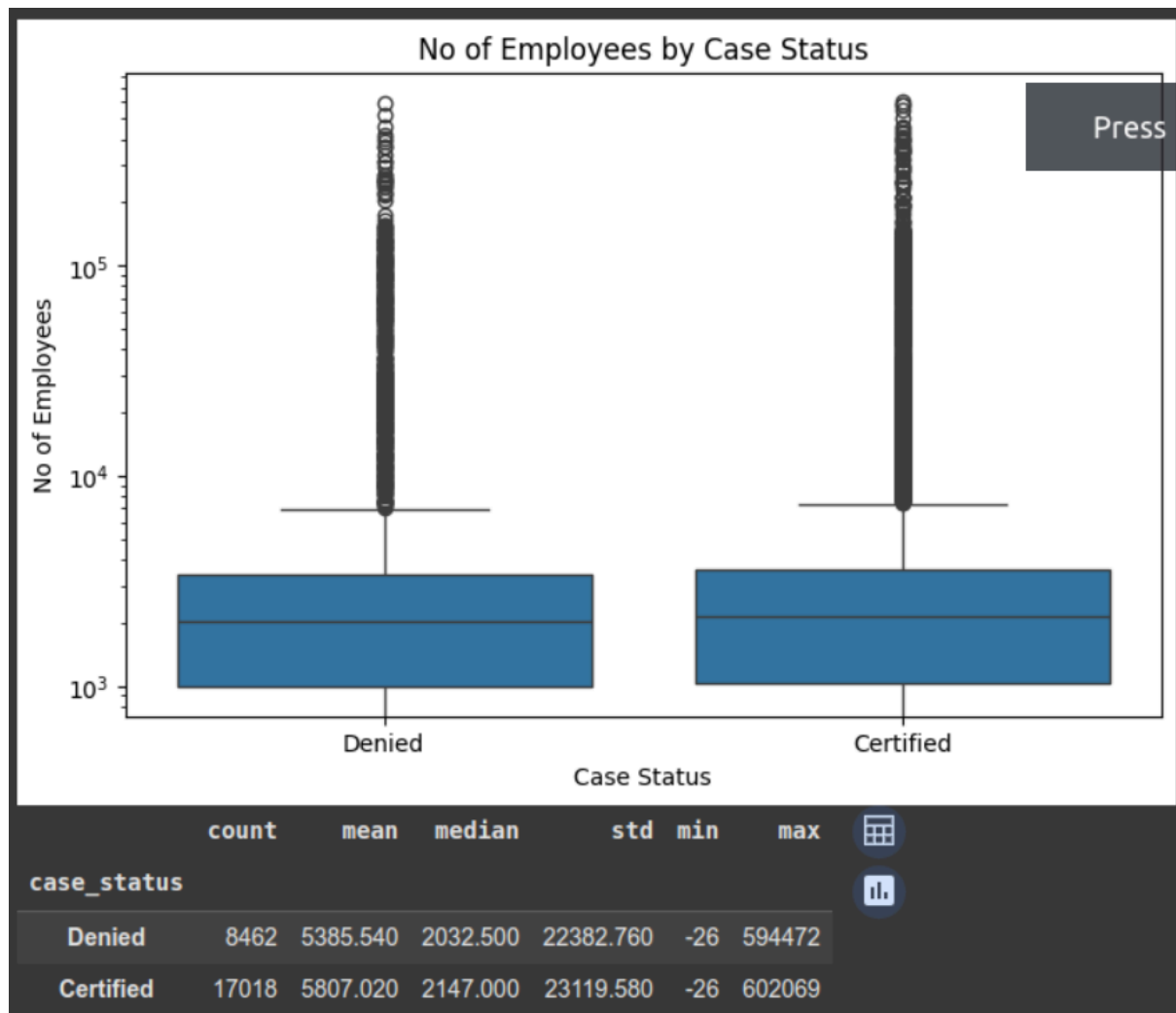


Fig.19: Number of Employees by Case Status

The **no_of_employees** variable represents the total workforce size of the employer applying for the visa. Larger organizations generally have more structured compliance systems, better documentation, and higher credibility in the eyes of labor authorities.

Observations

- The **average number of employees** is slightly higher for **certified cases** ($\approx 5,807$) compared to **denied cases** ($\approx 5,386$).
- The **median values** also show a modest difference — **2,147 employees** for certified vs **2,033** for denied.
- Both distributions are **heavily right-skewed**, with extreme outliers — some firms have more than **600,000 employees**, pointing to the

presence of multinational corporations.

- A few **negative values** (e.g., -26) appear, almost certainly representing **data entry errors**, which should be treated or removed during preprocessing.

Key Observations

- Visa approvals tend to be **slightly higher for employers with larger workforce sizes**, possibly because well-established firms are viewed as more financially stable and compliant with regulatory norms.
- However, the **difference between certified and denied cases is minor**, indicating that company size alone does not strongly determine visa outcomes.
- It is likely that **no_of_employees** interacts with other factors — such as **prevailing wage**, **region**, and **education level** — to affect certification probability.

Insight

Employers with **larger and more established operations** enjoy a modest advantage in visa approval rates, but **company size alone is not a strong predictor**. Before modeling, **data anomalies** (like negative employee counts) should be corrected or excluded to maintain data integrity.

Case Status by Education Level

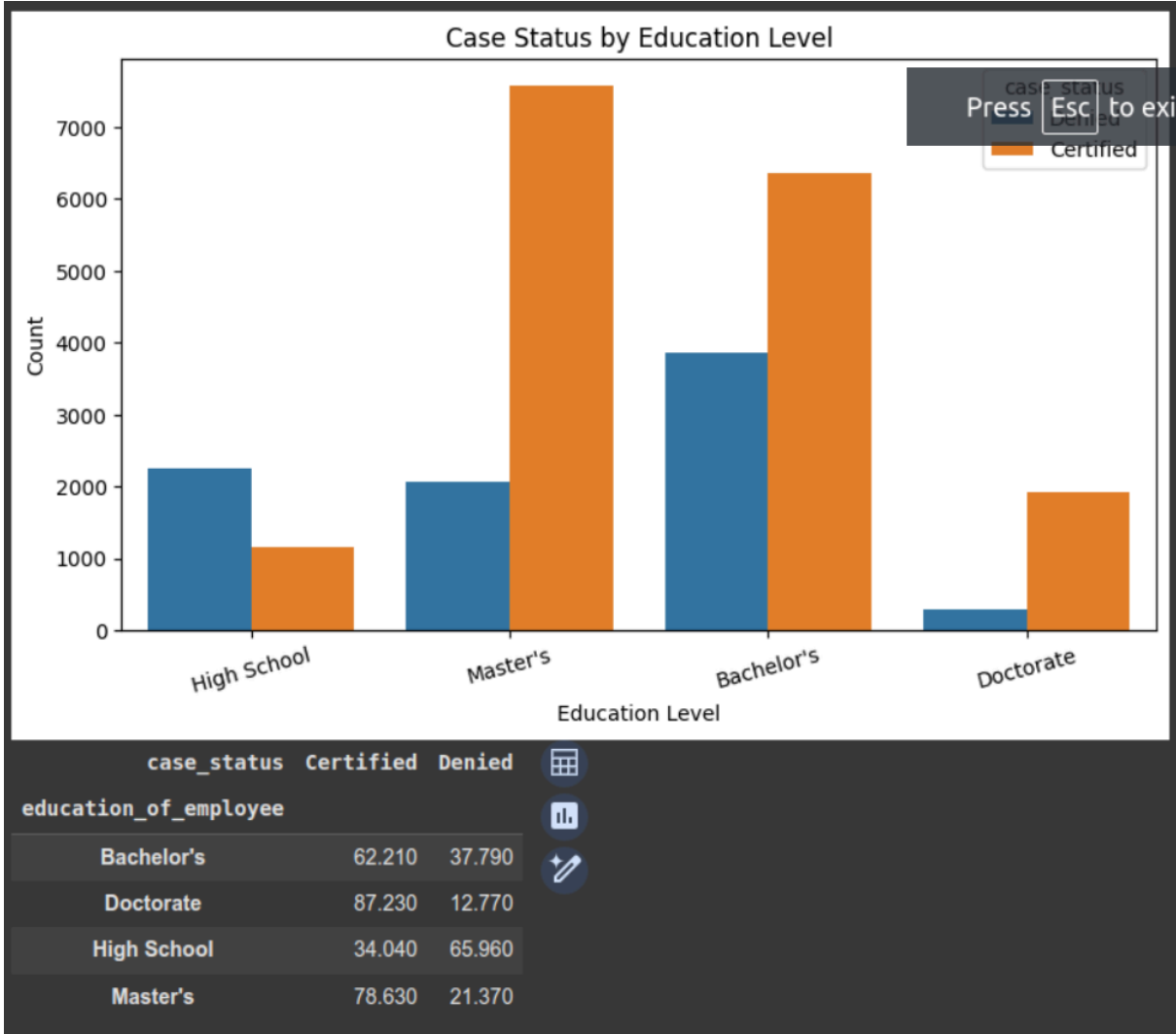


Fig.20: Case Status by Education Level

The **education_of_employee** variable captures the highest educational qualification of the visa applicant. Education is a key indicator of skill level and employability, making it a strong determinant of visa approval outcomes.

Observations

- **Visa certification rates** rise consistently with education level:
 - **Doctorate:** 87.2% certified
 - **Master's:** 78.6% certified
 - **Bachelor's:** 62.2% certified
 - **High School:** 34.0% certified
- Applicants with **lower educational attainment** face far higher denial rates — nearly **two-thirds of High School graduates** were denied.
- The relationship between education and approval is **monotonic**, meaning certification probability increases steadily with higher education.

Key Observations

- Higher education **substantially increases visa approval odds**, as employers can more easily justify the need for specialized foreign talent unavailable in the domestic labor pool.
- OFLC's approval patterns align with the U.S. immigration policy focus on **attracting high-skill professionals** in fields like technology, engineering, and research.
- Education likely interacts with **prevailing wage** and **job experience**, since applicants with advanced degrees often secure higher-paying, specialized roles.

Insight

Educational attainment is one of the **strongest predictors** of visa certification. Applicants with **Master's or Doctorate degrees** enjoy a clear advantage, reflecting the system's prioritization of **skilled, high-value immigration** that contributes to innovation and competitiveness in the U.S. economy.

Case Status by Continent

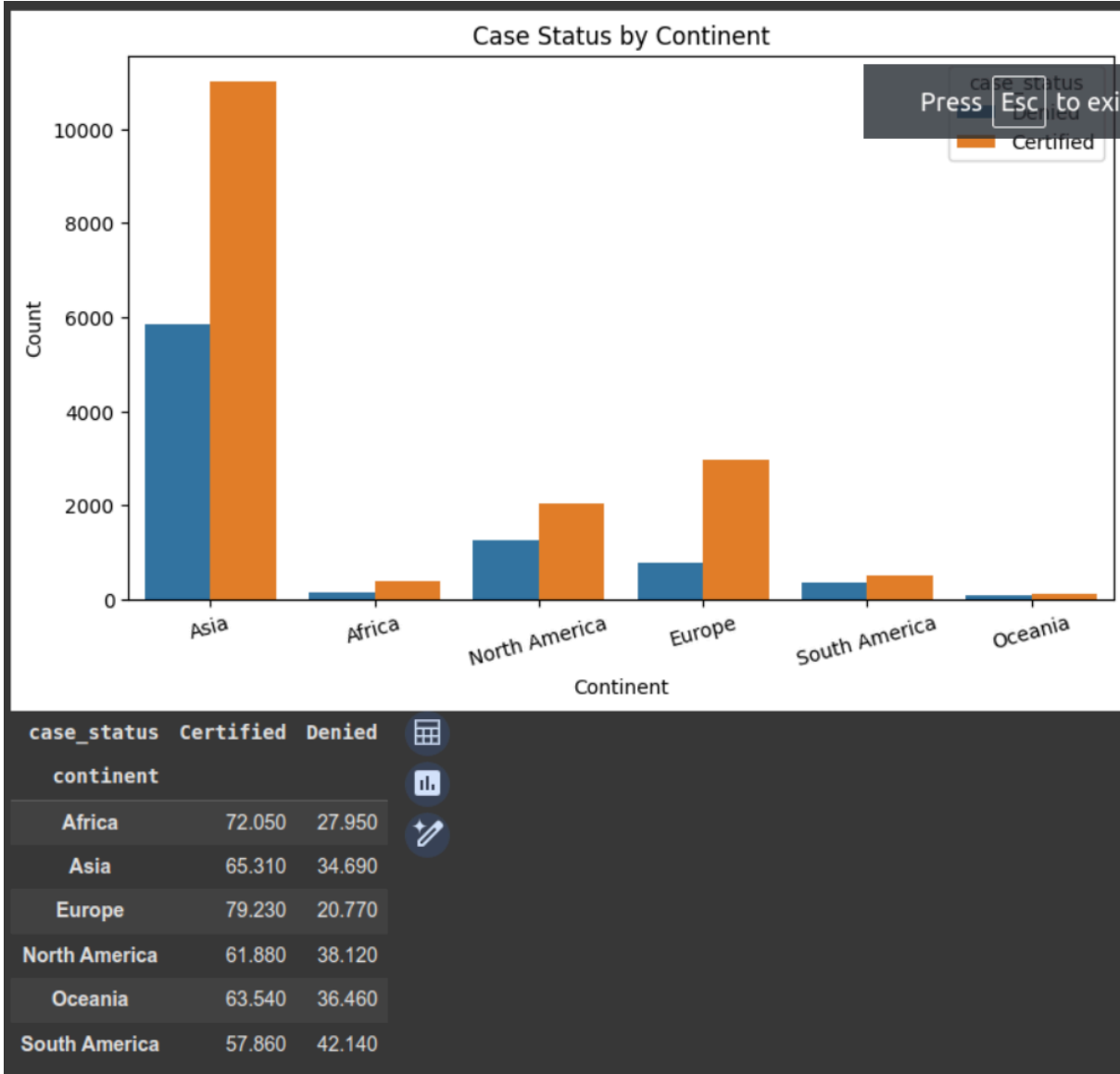


Fig.21:Case Status by Continent

The **continent** variable indicates the applicant’s continent of origin. It helps reveal regional patterns and potential disparities in visa approval rates, reflecting how labor demand and employer profiles vary globally.

Observations

- **Europe** has the highest visa approval rate at **79.2%**, followed by **Africa (72.1%)** and **Asia (65.3%)**.
- Applicants from **South America (57.9%)**, **Oceania (63.5%)**, and **North America (61.9%)** show comparatively lower certification rates.
- **Asia** dominates the dataset, contributing more than **two-thirds of all applications**, making it the primary driver of overall visa trends.

Key Observations

- Although Asian applicants form the majority, their **approval rate is only moderate**, implying more competition or stricter evaluation within this group.
- **Europe and Africa** demonstrate higher success rates, which might stem from stronger employer documentation, higher average wages, or occupations with greater labor demand alignment.
- The relatively **lower approval rates** for **South America and Oceania** could reflect smaller sample sizes or concentration in less specialized roles.

Insight

Visa approval outcomes show clear **regional disparities**. **European and African applicants** enjoy higher certification rates, possibly due to more compliant employers and well-matched job profiles. **Asian applicants**, despite being the largest applicant group, face **greater competition and scrutiny**, making continent a subtle but meaningful factor in predicting visa outcomes.

Case Status by Job Experience



Fig.22:Case Status by Job Experience

The **has_job_experience** variable captures whether the applicant has prior professional experience. Experience often signals employability, practical skill, and reduced training requirements—all of which are influential factors in visa approval outcomes.

Observations

- Applicants **with job experience (Y)** have a **notably higher approval rate of 74.5%**, compared to only **56.1%** for those **without experience (N)**.
- Correspondingly, **43.9%** of inexperienced applicants were denied, versus **25.5%** among experienced ones.
- The visual and tabular data confirm a **strong, consistent advantage** for experienced workers in visa certification outcomes.

Key Observations

- The OFLC's apparent preference for experienced applicants likely stems from their **immediate job readiness** and lower need for employer-sponsored training.
- Prior work experience also represents **lower perceived risk** for employers and regulators, implying smoother onboarding and higher productivity potential.
- The magnitude of difference between the two groups suggests that **experience is one of the strongest predictors** of certification success across all applicant features.

Insight

Having prior job experience dramatically improves visa approval chances. This reinforces the labor program's focus on **importing skilled, work-ready professionals** rather than entry-level or trainable candidates, emphasizing the U.S. system's orientation toward **immediate economic contribution**.

Case Status by Requirement of Job Training



Fig.23:Case Status by Requirement of Job Training

The **requires_job_training** variable reflects whether the offered position involves formal, employer-provided job training. This can indicate both the complexity of the role and the employer's willingness to invest in employee development.

Observations

- The **visa certification rate** for applicants **not requiring job training (N)** is **66.7%**, while for those **requiring training (Y)** it is slightly higher at **67.9%**.
- The **difference between the two groups is negligible**, showing that training requirements have little measurable influence on visa outcomes.
- Denial rates remain roughly similar ($\approx 32\text{--}33\%$) for both categories, implying consistent evaluation regardless of training needs.

Key Observations

- The data suggests that **requiring job training neither helps nor hinders** certification chances in any significant way.
- OFLC's decisions likely hinge more on **education, job experience, and prevailing wage**, rather than whether the position mandates formal training.
- Employers offering training may even be perceived as more invested in employee growth, which could subtly offset any potential disadvantage.

Insight

The requirement for **job training exerts minimal impact** on visa certification outcomes. The approval process seems guided more by **skill level, wage fairness, and employer reliability** than by the inclusion of training programs

Case Status by Region of Employment

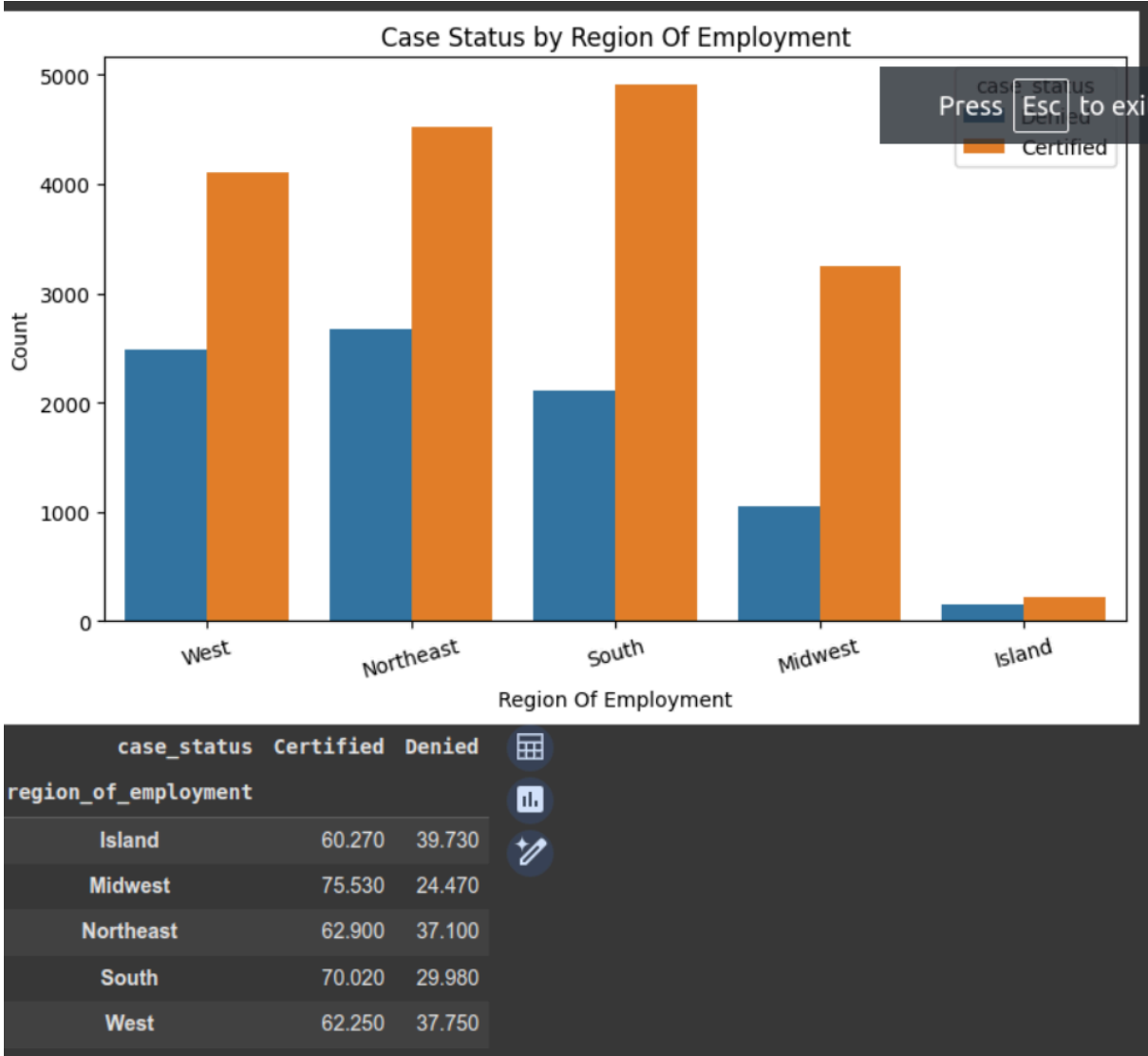


Fig.24:Case Status by Requirement of Job Training

The **region_of_employment** variable identifies the U.S. region where the sponsored employee is expected to work. Regional disparities in approval rates may reflect differences in local labor demand, wage norms, and regulatory scrutiny.

Observations

- The **Midwest** shows the highest certification rate at **75.5%**, followed by the **South (70.0%)**.
- The **Northeast (62.9%)** and **West (62.3%)** have moderate approval rates, while **Island regions** (e.g., Hawaii, Puerto Rico) record the lowest at **60.3%**.
- Correspondingly, denial rates are **higher in the West and Northeast**, suggesting either more competitive job markets or more stringent evaluations in those areas.

Key Observations

- The **Midwest and South** likely lead in approvals because of strong demand in sectors like **manufacturing, agriculture, logistics, and energy**, which rely heavily on foreign labor and often face chronic domestic worker shortages.
- Regions such as the **West and Northeast**, with higher native employment and concentrations of white-collar industries, may experience **greater competition** and **stricter scrutiny** of visa applications.
- The **Island region's** small sample size makes its results less reliable and more prone to statistical noise.

Insight

Visa approval outcomes clearly vary by region. The **Midwest and South** demonstrate more favorable approval trends, reflecting regional labor gaps and policy flexibility, while the **Northeast and West**—characterized by dense urban labor markets—face tougher evaluations and relatively higher denial rates. This pattern underscores how **regional economic structures and workforce dynamics** influence visa certification probabilities.

Case Status by Unit of Wage

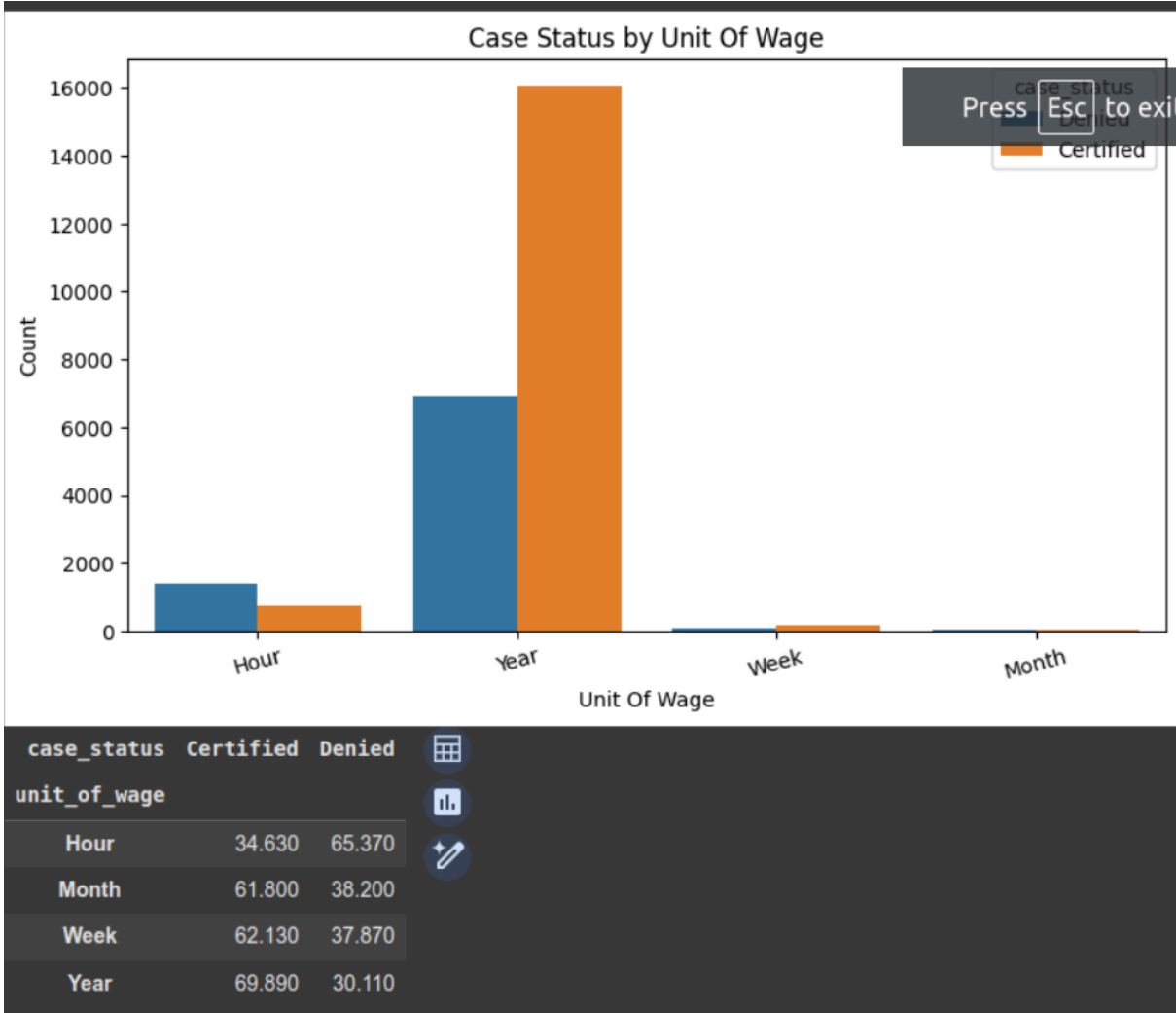


Fig.25:Case Status by Unit of Wage

The **unit_of_wage** variable describes how the employee’s compensation is structured — hourly, weekly, monthly, or yearly. This reflects both the **nature and stability of employment**, factors that can significantly influence visa certification outcomes.

Observations

- Applicants **paid yearly** have the highest approval rate at **69.9%**, followed by **weekly (62.1%)** and **monthly (61.8%)** categories.
- **Hourly wage** applicants have the **lowest approval rate at 34.6%**, with **65.4% denied**.
- Most applications fall under the **yearly category**, consistent with **full-time, salaried professional roles**.

Key Observations

- The stark difference in approval rates shows that **salaried, full-time employment** is strongly favored in the certification process.
- **Hourly wage positions**, typically associated with **temporary, part-time, or lower-skilled work**, face higher scrutiny and rejection—likely due to OFLC’s emphasis on fair wage compliance and prevention of labor exploitation.
- Weekly and monthly wage categories represent a small minority of applications and show **moderate approval rates**, but the limited sample size makes these results less conclusive.

Insight

Visa certification outcomes are **closely linked to how wages are structured**. Positions offering **yearly salaries**—a proxy for stability and long-term employment—are far more likely to be approved. In contrast, **hourly wage jobs**, often viewed as less secure or lower-skilled, show the **lowest probability of certification**, reaffirming OFLC’s preference for **stable, high-value employment relationships**.

Case Status by Full-Time Position

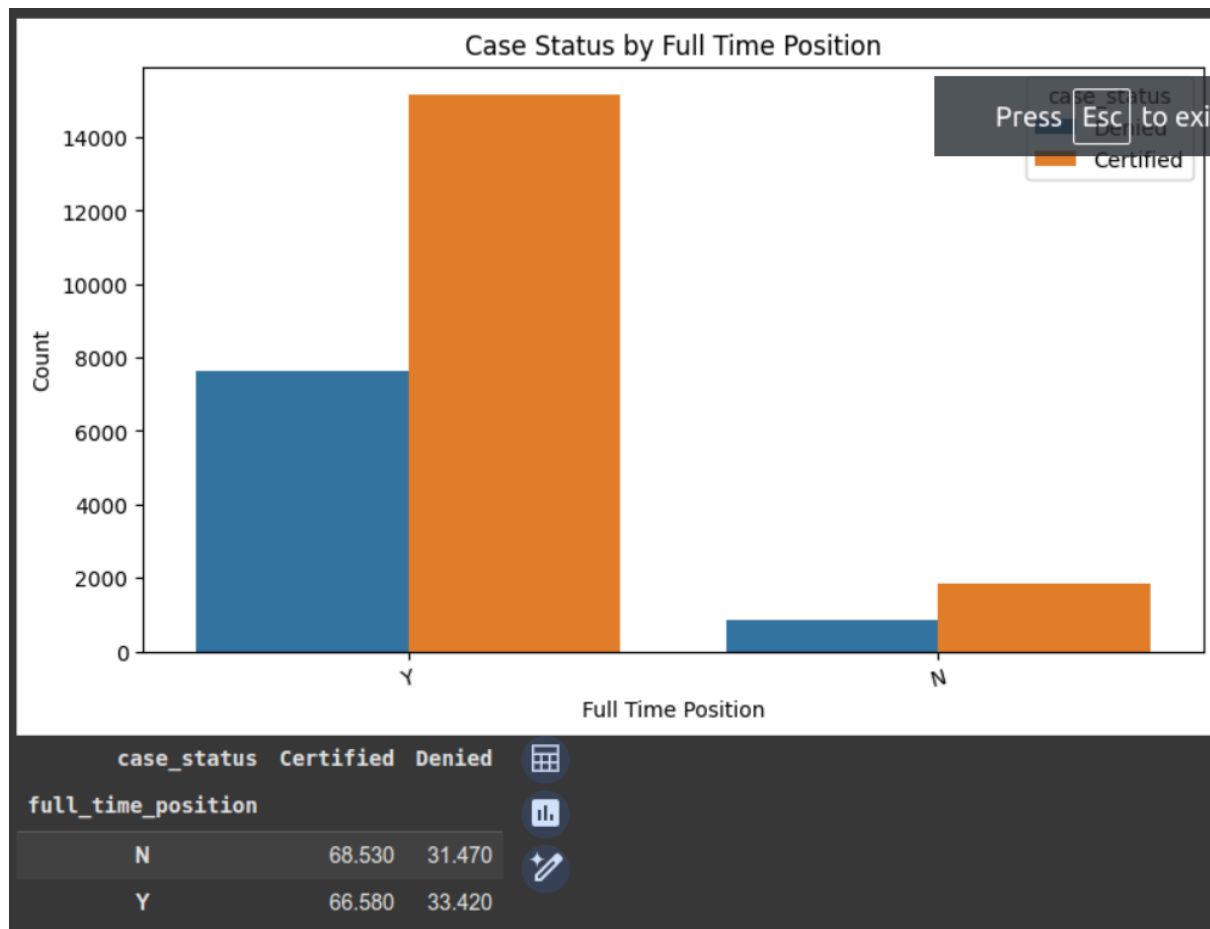


Fig.26:Case Status by Full-Time Position

The **full_time_position** variable specifies whether the offered role is **full-time** or **part-time**. Full-time employment is typically perceived as more stable and economically valuable, so one might expect it to yield higher visa certification rates—but the data tells a subtler story.

Observations

- The **visa certification rate** for **full-time positions (Y)** is **66.6%**, nearly identical to **part-time positions (N)** at **68.5%**.
- The difference between the two categories is **statistically negligible**, indicating that the **nature of employment (full-time vs. part-time)** does not play a decisive role in visa outcomes.
- Nonetheless, the overwhelming majority of applications—close to **90%**—are for full-time positions, making them the dataset's dominant category.

Key Observations

- The similar approval rates suggest that **OFLC decisions are not biased toward full-time roles** once wage and compliance criteria are met.
- This finding implies that the agency focuses primarily on **fair compensation, job legitimacy, and applicant qualifications**, rather than contract duration or weekly work hours.
- The small sample size of part-time positions may limit the statistical strength of this comparison, but the pattern still holds consistently.

Insight

Visa certification appears **largely independent of employment type**. As long as employers meet **prevailing wage and labor condition standards, part-time and full-time positions** are evaluated on equal terms—underscoring OFLC’s focus on **job quality and regulatory compliance over employment hours**.

Case Status by Full-Time Position

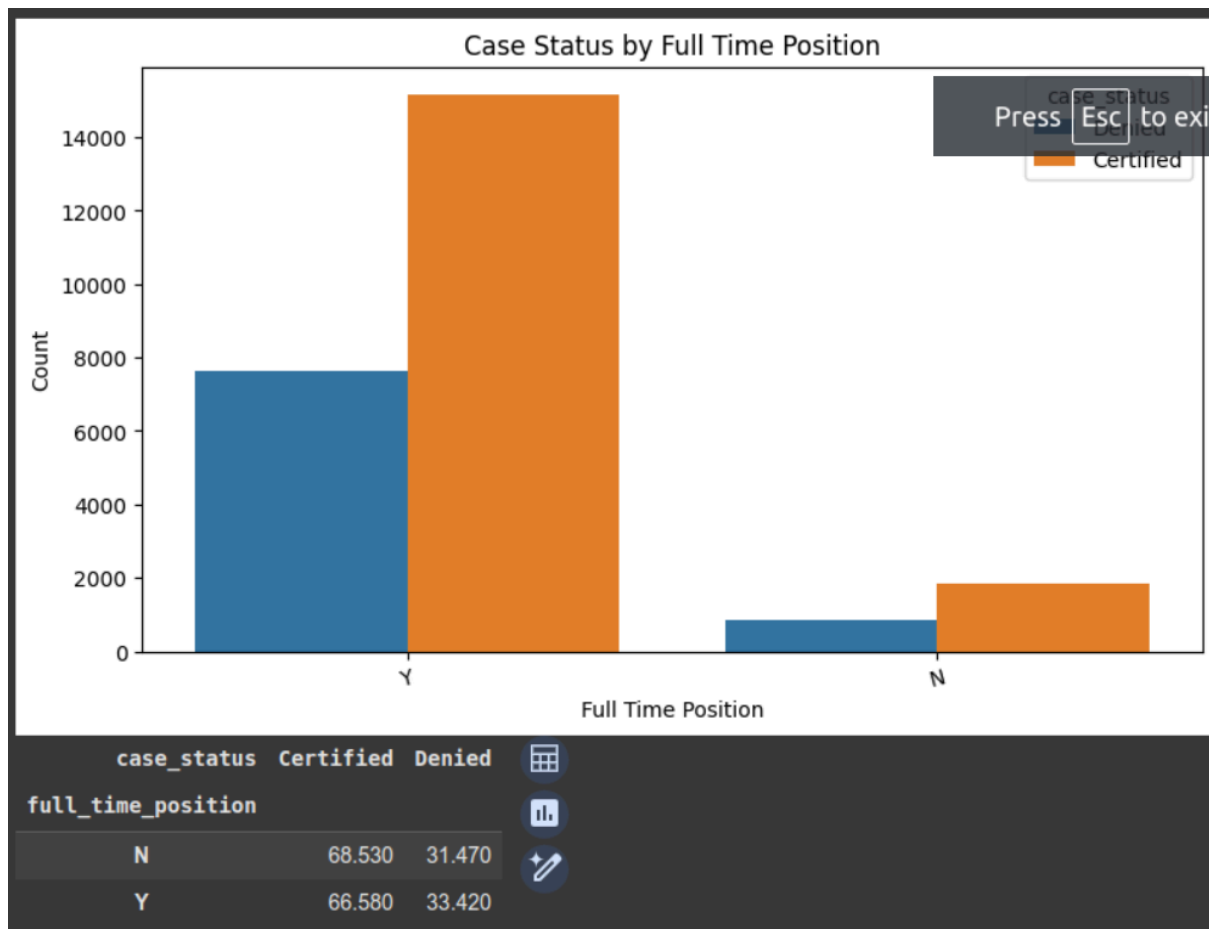


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- Nonetheless, the overwhelming majority of applications—close to **90%**—are for full-time positions, making them the dataset's

Key Insights based on EDA

1. Higher Education Drives Approval

Visa certification rates increase with education — **Doctorates (87%)** and **Master's (79%)** dominate approvals, while **High School** applicants face the highest denials. Education is the single strongest predictor of visa success.

2. Job Experience Greatly Boosts Success Rates

Applicants **with prior experience** have a **74.5% approval rate**, compared to **56.1%** for inexperienced ones. Experienced candidates are seen as job-ready and lower risk for employers.

3. Higher Wages Correlate with Higher Approvals

Certified applications have **~\$8,500 higher average prevailing wages** than denied ones. Offering competitive pay improves approval odds, signaling fair labor practices and specialized skill sets.

4. Regional Differences Affect Certification

Approval is highest in the **Midwest (75.5%)** and **South (70%)**, but lower in the **Northeast (62.9%)** and **West (62.3%)**. This suggests labor shortages and demand variation across U.S. regions influence visa outcomes.

5. Company Size and Age Have Limited Impact

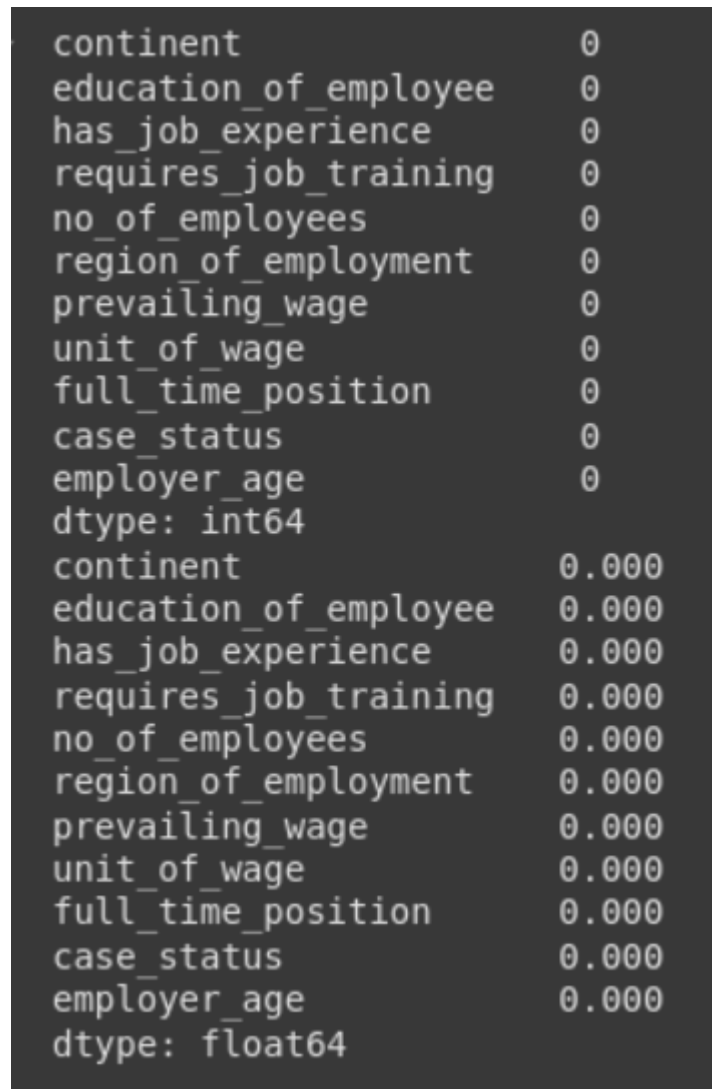
Larger, older companies show only a **slight edge** in approval rates. Visa outcomes depend more on wage and compliance factors than firm size or longevity.

6. Employment Type Has Minimal Effect

Full-time and part-time roles have **nearly identical approval rates (~67%)**, indicating OFLC evaluates applications primarily on **qualification and wage fairness**, not job duration.

Data Preprocessing

Data Quality Checks



continent	0
education_of_employee	0
has_job_experience	0
requires_job_training	0
no_of_employees	0
region_of_employment	0
prevailing_wage	0
unit_of_wage	0
full_time_position	0
case_status	0
employer_age	0
dtype: int64	
continent	0.000
education_of_employee	0.000
has_job_experience	0.000
requires_job_training	0.000
no_of_employees	0.000
region_of_employment	0.000
prevailing_wage	0.000
unit_of_wage	0.000
full_time_position	0.000
case_status	0.000
employer_age	0.000
dtype: float64	

Fig.27:Missing values

Before advancing to outlier detection or model development, it's vital to confirm that the dataset is accurate, consistent, and free from integrity issues. Two key checks were conducted: **missing value analysis** and **duplicate row detection**.

1. Missing Value Analysis

To assess data completeness, each column was examined for null or missing entries. Both the absolute count and the percentage of missing values were computed using:

Result:

All variables—numeric and categorical—contained **0 missing values (0%)**.

This confirms that the dataset is **fully complete**, eliminating the need for imputation or record removal. Every observation is ready for subsequent preprocessing and modeling.

2. Duplicate Row Analysis

Duplicate entries can distort model training and misrepresent feature distributions. To ensure record uniqueness, the following check was performed:

Result:

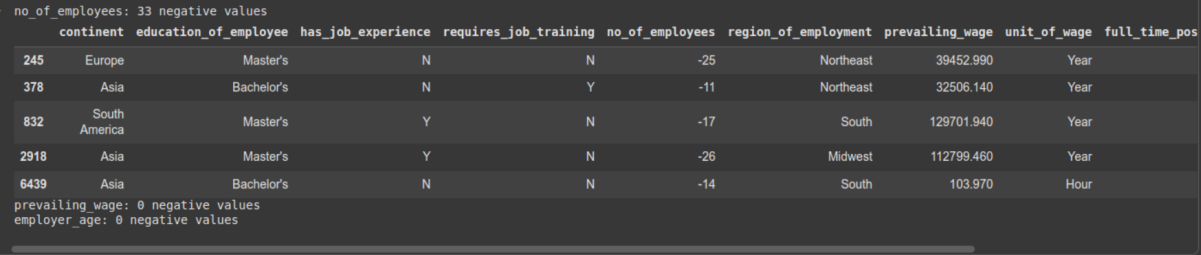
The dataset contained **0 duplicate rows**, verifying that **each record is unique**. This confirms the integrity and reliability of the dataset for downstream analyses.

Conclusion:

The dataset passes both **completeness** and **uniqueness** checks, ensuring a **clean foundation** for further exploratory analysis, feature engineering, and model building.

3. Handling Impossible Negative Values

During data validation, the numeric variables were checked for **impossible negative entries**—values that are not logically possible in real-world terms (e.g., negative number of employees or company age).



```
no_of_employees: 33 negative values
continent education_of_employee has_job_experience requires_job_training no_of_employees region_of_employment prevailing_wage unit_of_wage full_time_pos
245 Europe Master's N N -25 Northeast 39452.990 Year
378 Asia Bachelor's N Y -11 Northeast 32506.140 Year
832 South America Master's Y N -17 South 129701.940 Year
2918 Asia Master's Y N -26 Midwest 112799.460 Year
6439 Asia Bachelor's N N -14 South 103.970 Hour
prevailing_wage: 0 negative values
employer_age: 0 negative values
```

Fig.28:Impossible negatives in no_of_employees

Detection:

The check revealed that the variable **no_of_employees** contained a few negative values (for example, -26), which are not realistic. Other numeric variables, including **prevailing_wage** and **employer_age**, showed no such issues.

Treatment:

Since negative employee counts are invalid, those records were identified as **data-entry errors**. The incorrect values were removed to preserve the integrity and reliability of the dataset.

Verification:

A follow-up review confirmed that no numeric variable contained any negative values after cleaning.

Key Takeaway:

Eliminating impossible values ensures the dataset reflects real-world conditions, strengthening its **credibility**, improving **data consistency**, and enhancing the **accuracy** of all subsequent analyses and modeling.

Outlier Detection

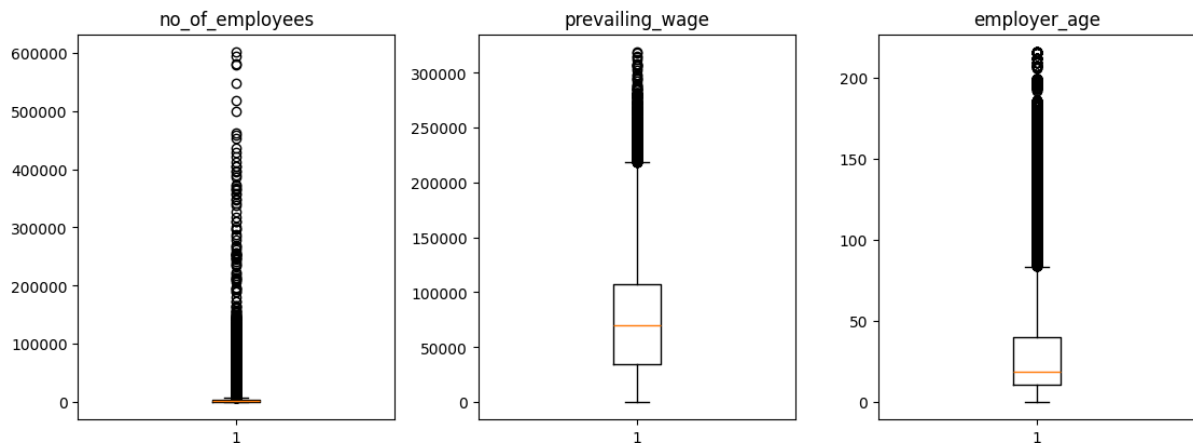


Fig.29: Outlier Detection

Outlier detection was carried out for all numeric variables to identify data points that deviate substantially from the rest of the distribution. Boxplots were used as the primary visual tool, applying the **Interquartile Range (IQR)** rule, where any observation lying beyond $1.5 \times \text{IQR}$ from the lower or upper quartile is classified as a potential outlier.

Approach

Boxplots were examined for the following numeric variables:

- no_of_employees
- prevailing_wage
- employer_age

This analysis provided a clear visual understanding of the data spread and helped pinpoint extreme values that might distort model performance or bias summary statistics.

Findings

- **No. of Employees:**

The distribution is **heavily right-skewed**, with a cluster of very large firms creating a long tail. Some organizations reported more than **600,000 employees**—plausible for multinational corporations but extreme relative to most employers. Such large values may skew statistical results and need scaling or capping.

- **Prevailing Wage:**

The data displays a pronounced **right-skew**, with a subset of applications reporting wages above **\$300,000**. These cases likely correspond to executive or highly specialized technical roles.

Although valid, such extremes can heavily influence model coefficients and will benefit from transformation (e.g., log scaling).

- **Employer Age:**

Most companies fall within the **10–50 year** range, which reflects normal corporate lifespans. However, a small number exceed **150 years**, which is unlikely to be realistic and may reflect **data entry errors** rather than genuine observations. These will require correction or exclusion.

Conclusion

Outliers were detected in all three numeric variables, particularly in **no_of_employees** and **prevailing_wage**. While some outliers may represent genuine business cases, others—especially unrealistic company ages—must be addressed to prevent model distortion.

In subsequent preprocessing, these outliers will be **treated through capping, transformation, or removal**, ensuring the dataset remains

statistically reliable and that the final model is both **robust and interpretable**.

Outlier Treatment (Log Transformation Results)

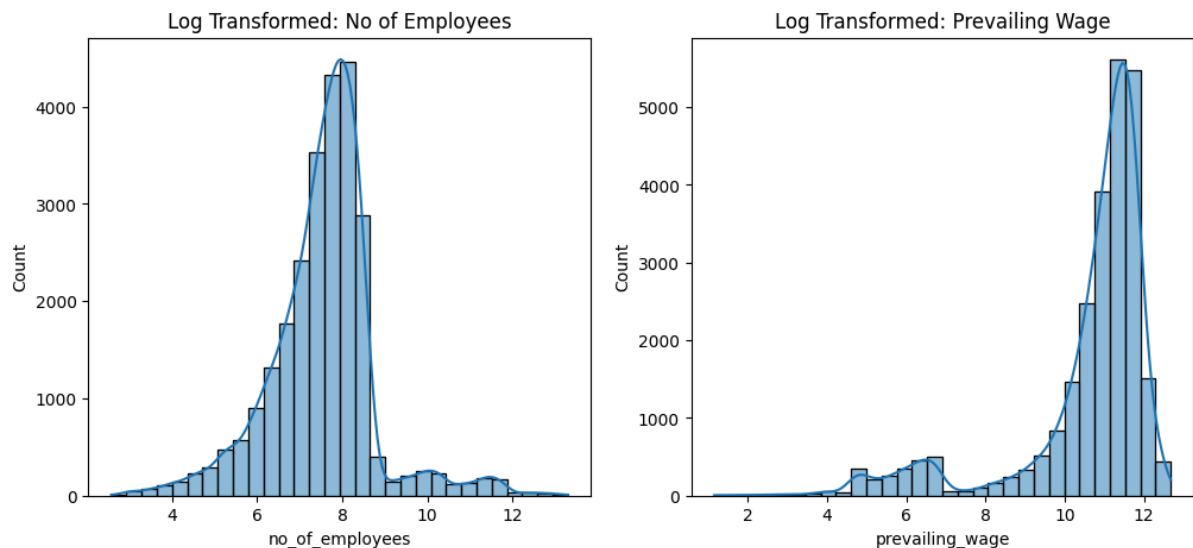


Fig.30: Outlier Treatment (Log Transformation Results)

After applying the natural log transformation using `np.log1p()` on **no_of_employees** and **prevailing_wage**, both variables exhibited a substantial improvement in distribution shape.

The original versions were **highly right-skewed**, dominated by a few extreme values that stretched the scale and distorted variance. The log transformation effectively **compressed large magnitudes**, reducing the impact of outliers while maintaining relative differences between observations.

The post-transformation histograms reveal much smoother, more symmetric patterns:

- **no_of_employees** now closely resembles a normal distribution with significantly reduced spread.
- **prevailing_wage** retains a mild multimodal structure—reflecting variations in wage tiers—but displays much less skewness and improved overall balance.

This transformation enhances model interpretability and numerical stability by **mitigating the disproportionate influence of extreme values**. The dataset is now better conditioned for robust **statistical analysis and machine learning modeling**.

Outlier Treatment – Employer Age

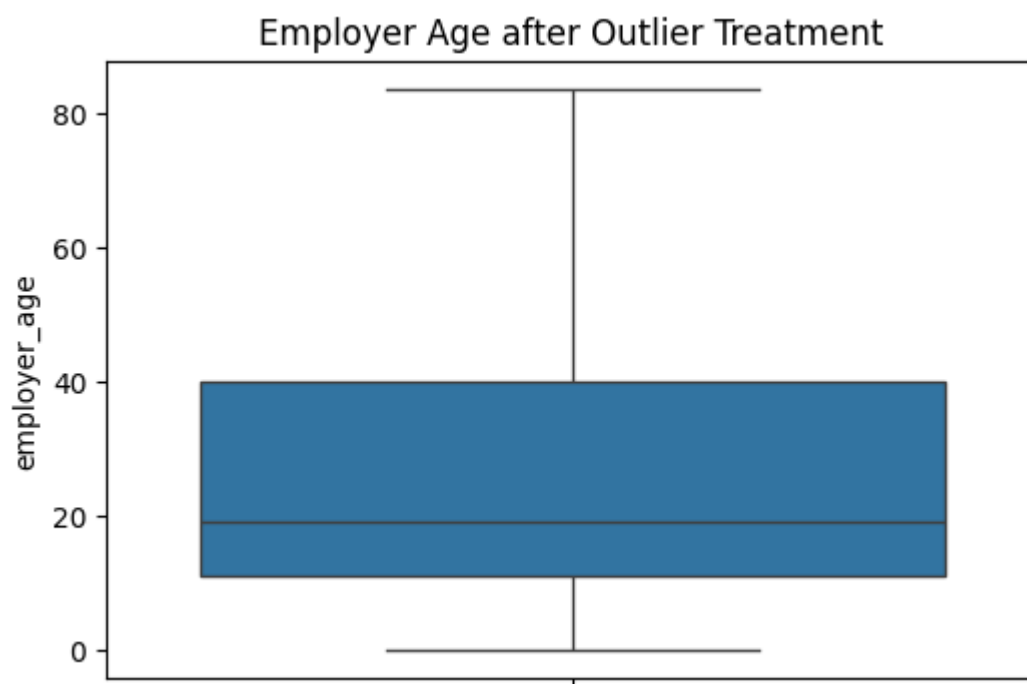


Fig.31: Outlier Treatment – Employer Age

Outlier detection for **employer_age** revealed several unrealistic values, with some companies appearing to be more than **200 years old**—an impossibility in practical terms. To address this, the **Interquartile Range (IQR) method** was used for capping extreme values rather than removing them, preserving data volume while enhancing reliability.

The **upper threshold** was computed as $Q3 + 1.5 \times IQR$, and any values exceeding this boundary were **replaced with the upper-limit value**. This approach ensures that extreme but valid older firms are retained, while clearly erroneous entries are corrected.

After capping, the **maximum employer age** dropped to approximately **85 years**, producing a far more plausible and stable distribution. The updated boxplot confirmed the removal of extreme outliers, resulting in a dataset that is **cleaner, more realistic, and better aligned with business logic**.

Feature Engineering

Data Splitting and Encoding

To prepare the data for modeling, the dataset was divided into **training, validation, and test sets** in an **80/5/15 ratio**. The validation set was taken from a temporary holdout portion of the data. The split was **stratified by the target variable (case_status)**, meaning that the proportion of certified and denied cases was kept consistent across all three subsets. This ensures that each subset reflects the same balance of approvals and denials as the full dataset, preventing bias during model training and evaluation.

Next, all **categorical predictor variables** were converted into numeric form through **one-hot encoding**—a method that creates binary indicator columns for each category within a variable. For example, if a feature had three categories (A, B, and C), it would be represented by two new columns (B and C), with A serving as the baseline category. This avoids redundancy and ensures clear numerical representation of categorical information.

Because some categories may not appear in the smaller validation or test subsets, those subsets were later **aligned with the training feature set** so that all three datasets contained the same columns in the same order. Any missing category columns were filled with zeros, which is appropriate since a zero simply means that a given category was not present for that record. This step guarantees that the model encounters the same structure during training and testing, eliminating feature mismatch issues.

Finally, the **target variable** (**case_status**) was converted from text labels into numerical form:

0 representing *Denied* and **1** representing *Certified*.

This transformation standardizes the output for all machine learning algorithms and simplifies the calculation of performance metrics such as accuracy, recall, precision, and F1-score.

Model building

Model Building - Original Data

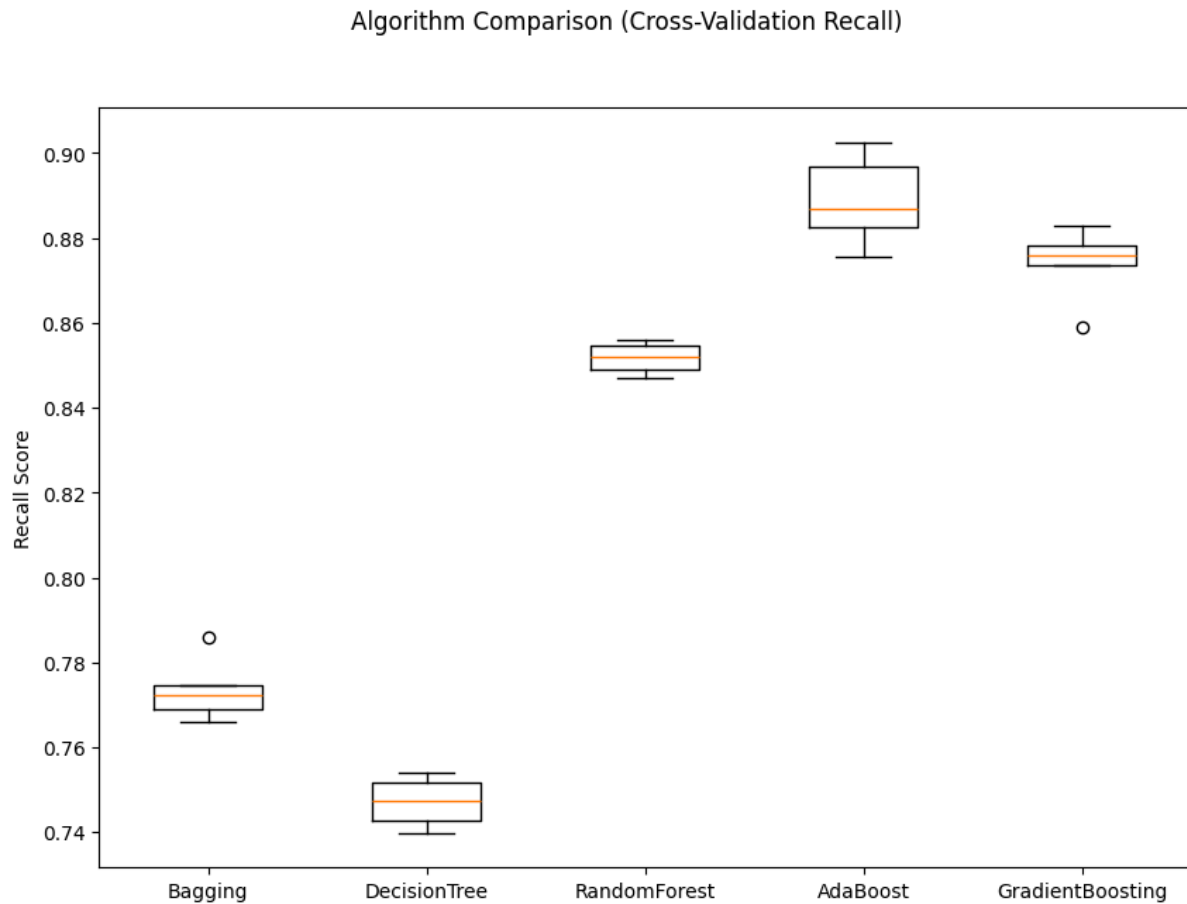


Fig.32: Algorithm Comparison (Cross-Validation Recall) on Original Data

After preprocessing and data cleaning, several ensemble and tree-based classification algorithms were trained on the original dataset to predict **case_status (Certified / Denied)**. The primary evaluation metric chosen was **Recall**, as it measures how effectively the model identifies *Certified* cases (minimizing false negatives).

Models Evaluated

- 1. **Bagging Classifier**
- 2. **Decision Tree Classifier**
- 3. **Random Forest Classifier**
- 4. **AdaBoost Classifier**
- 5. **Gradient Boosting Classifier**

Each model was evaluated using **5-fold Stratified Cross-Validation** on the training data. The Recall metric was computed for each fold, and the mean $\pm$ standard deviation were used to summarize performance stability. Validation Recall was also computed on the hold-out validation set to assess generalizatio

Cross-Validation (Recall) Results		
Model	Mean Recall	Std Dev
Bagging	0.7736	$\pm$ 0.0069
Decision Tree	0.7472	$\pm$ 0.0053
Random Forest	0.8516	$\pm$ 0.0034
AdaBoost	0.8887	$\pm$ 0.0097
Gradient Boosting	0.8738	$\pm$ 0.0081

Validation Recall Scores	
Model	Validation Recall
Bagging	0.7345
Decision Tree	0.7467
Random Forest	0.8230
AdaBoost	0.8473
Gradient Boosting	0.8388

Interpretation

- **AdaBoost** achieved the **highest mean recall (0.8887)** during cross-validation and also performed strongly on the validation set (**0.8473**), indicating excellent sensitivity and stability.
- **Gradient Boosting** followed closely with competitive recall both in CV and validation.
- **Random Forest** performed consistently well but slightly below the boosting algorithms.
- **Bagging** and **Decision Tree** showed lower recall and higher variance, suggesting limited generalization capability.

Visualization

The following boxplot compares the distribution of Recall scores across the 5 cross-validation folds for all models:

(Refer to the figure above — “Algorithm Comparison (Cross-Validation Recall)”)

From the boxplot, **AdaBoost and Gradient Boosting** exhibit higher median recall and narrower spread, confirming their superior and more stable performance across folds.

Conclusion:

The above results are based on the original (imbalanced) dataset, where the minority class (“Denied” cases) is underrepresented.

Among all models, AdaBoost achieved the highest recall (0.8887 in cross-validation, 0.8473 on validation), followed closely by Gradient Boosting. These boosting methods outperformed Bagging and single Decision Trees by a significant margin, demonstrating their ability to capture complex patterns even in imbalanced data.

However, since the data is imbalanced, these recall scores may still favor the majority class.

To further validate model robustness and ensure fair learning across both classes, the next step involves applying oversampling techniques (e.g., SMOTE or RandomOverSampler) and retraining the same models. This will allow a direct performance comparison between models trained on the original versus balanced datasets.

Model Building on Oversampled Data (SMOTE)

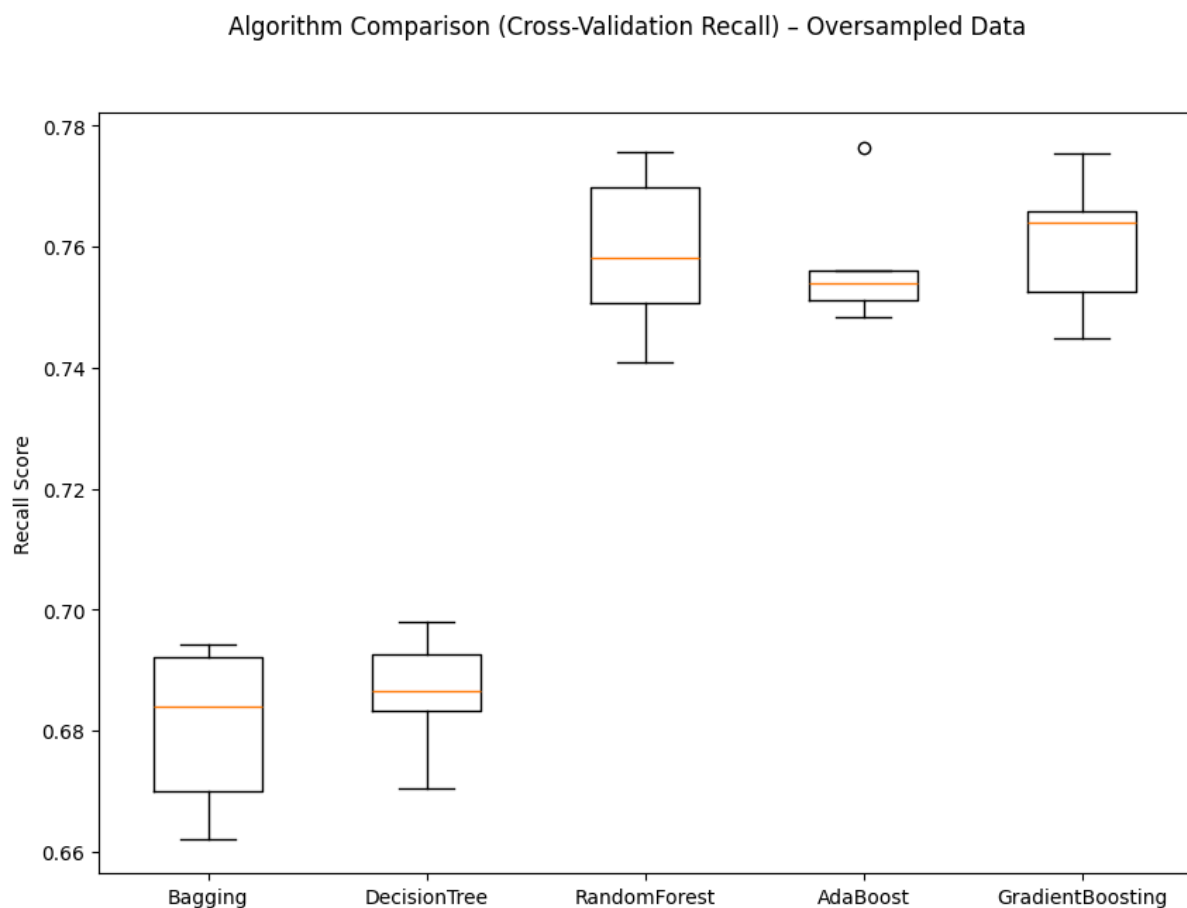


Fig.33: Algorithm Comparison (Cross-Validation Recall) on Oversampled Data

To address the imbalance between the “Certified” and “Denied” cases in the target variable, the **Synthetic Minority Oversampling Technique (SMOTE)** was applied to the training dataset. SMOTE generates synthetic samples of the minority class by interpolating between existing instances, creating a balanced dataset that allows models to better learn decision boundaries for both classes.

After oversampling, both classes had equal representation, reducing the bias toward the majority class and improving the model’s ability to recognize minority class instances.

Models Trained and Evaluation Approach

The same ensemble models used in the previous experiment on the original dataset were evaluated on the oversampled data. These included:

- **Bagging Classifier**
- **Decision Tree Classifier**
- **Random Forest Classifier**
- **AdaBoost Classifier**
- **Gradient Boosting Classifier**

Each model was evaluated using **5-fold Stratified Cross-Validation**, with **Recall** as the key performance metric. Recall was chosen because the objective of this analysis is to maximize the detection of “Certified” cases and minimize false negatives.

Cross-Validation Results (Recall)		
Model	Mean CV Recall	Std Dev
Bagging	0.6805	± 0.0125
Decision Tree	0.6862	± 0.0093
Random Forest	0.7591	± 0.0126
AdaBoost	0.7571	± 0.0100
Gradient Boosting	0.7605	± 0.0106

Validation Set Performance (Recall)	
Model	Validation Recall
Bagging	0.6364
Decision Tree	0.6461
Random Forest	0.7261
AdaBoost	0.7394
Gradient Boosting	0.7297

Interpretation

After oversampling, all models showed a slight decline in recall compared to the original data. This reduction suggests that oversampling introduced some **synthetic variance**, which may have made the models less generalizable to unseen data. However, the overall recall levels remain consistent across most ensemble methods.

Among all models, **AdaBoost (recall ≈ 0.74)** and **Gradient Boosting (recall ≈ 0.73)** performed the best, maintaining strong recall and stable variance across folds. **Random Forest** followed closely with a recall of approximately 0.72. **Bagging** and **Decision Tree** classifiers continued to

show lower recall scores, consistent with earlier observations on the original data.

The boxplot visualization comparing model recall across folds shows that **AdaBoost** and **Gradient Boosting** achieved the most consistent performance, indicating their robustness and stability in capturing the minority class after oversampling.

Model Building – Undersampled Data

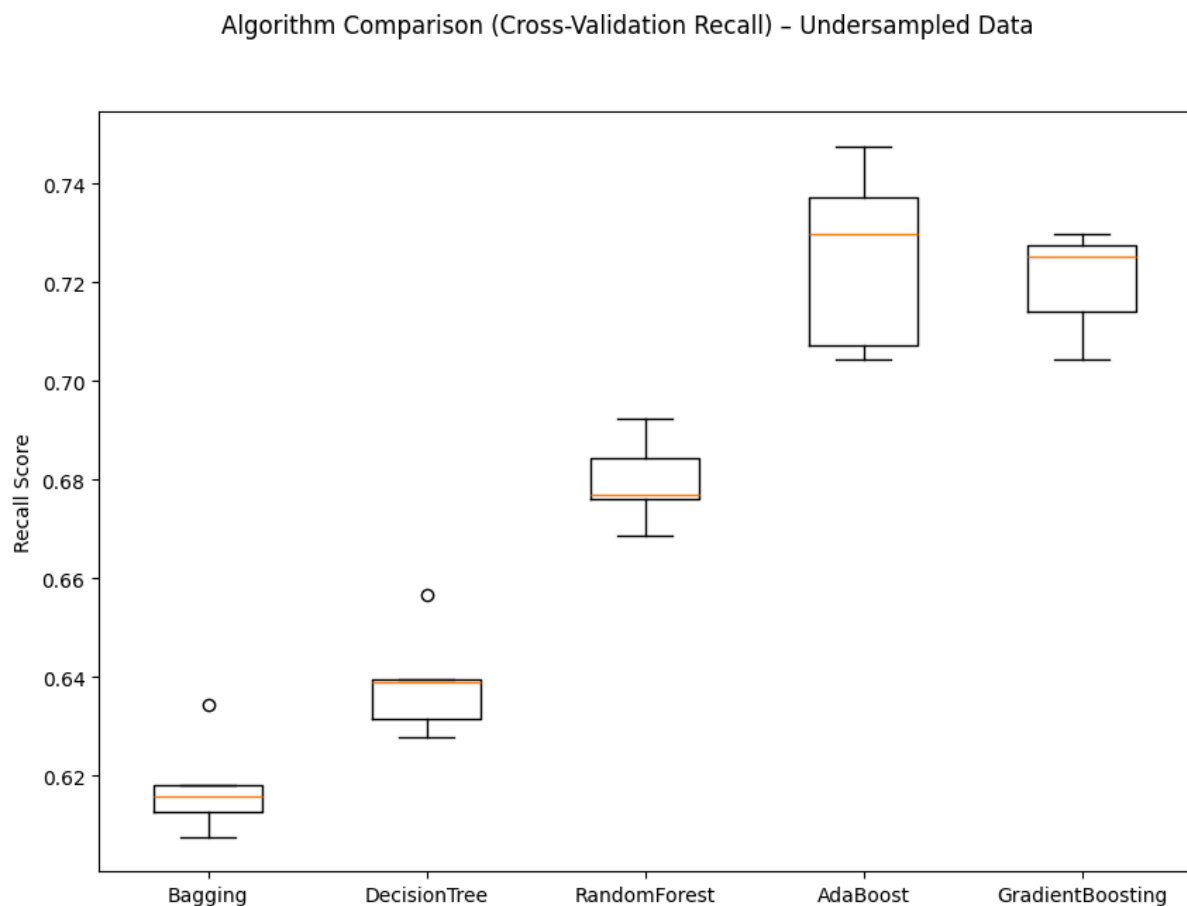


Fig.34: Algorithm Comparison (Cross-Validation Recall) on Undersampled data

To further assess how sampling techniques impact model performance, **random undersampling** was applied to balance the class distribution by reducing the number of majority class instances. Before undersampling, there were **13,642 Certified (Yes)** and **6,715 Denied (No)** cases, indicating a moderate imbalance. After applying the `RandomUnderSampler`, both classes were balanced to **6,715 instances each**.

Following undersampling, five ensemble and tree-based classification models were trained and evaluated using **5-fold Stratified Cross-Validation**, focusing on **Recall** as the primary performance metric (to minimize false negatives and ensure Certified cases are correctly identified

Cross-Validation Results (Recall)	
Model	Mean CV Recall
Bagging	0.6177
Decision Tree	0.6389
Random Forest	0.6797
AdaBoost	0.7252
Gradient Boosting	0.7202

Validation Recall	
Model	Validation Recall
Bagging	0.5624
Decision Tree	0.6073
Random Forest	0.6303
AdaBoost	0.6873
Gradient Boosting	0.6909

Interpretation

From the results and the algorithm comparison boxplot, **AdaBoost** and **Gradient Boosting** performed best in terms of recall on both cross-validation and validation sets. However, compared to the **original data** and **SMOTE oversampled data**, recall scores on undersampled data were noticeably lower across all models.

This reduction is expected since undersampling **discards useful information from the majority class**, leading to **reduced generalization capability**. Despite this, the models trained on undersampled data were able to maintain stable performance patterns

with boosting algorithms once again outperforming bagging and single decision trees.

Conclusion

From the results and the algorithm comparison boxplot, AdaBoost and Gradient Boosting performed best in terms of recall on both cross-validation and validation sets. However, compared to the original data and SMOTE oversampled data, recall scores on undersampled data were noticeably lower across all models. This reduction is expected since undersampling discards useful information from the majority class, leading to reduced generalization capability.

In conclusion, while undersampling balances the dataset and speeds up training, it comes at the cost of information loss and slightly weaker model performance compared to oversampling methods.

HyperParameter Tuning

Tuning AdaBoost using Undersampled Data

To improve the model's ability to correctly identify *Certified* cases while minimizing false negatives, AdaBoost was trained on **randomly undersampled data**. This balancing technique equalized the number of *Certified* and *Denied* cases, ensuring that the model learned decision boundaries more effectively across both classes.

A **RandomizedSearchCV** approach was used for hyperparameter tuning, with **5-fold cross-validation** and **F1-macro** as the optimization metric to balance precision and recall across classes. The parameter

search space included variations in the number of estimators, learning rate, and the maximum depth of the base decision tree classifier.

Best Parameters Identified:

- **n_estimators:** 90
- **learning_rate:** 0.2
- **Base estimator:** DecisionTreeClassifier(max_depth=3, random_state=1)

These parameters provided the highest cross-validated F1-macro score, indicating the most effective combination of complexity and generalization capability.

Model Performance						
Dataset	Accuracy	Balanced Accuracy	Precision (Binary)	Recall (Binary)	F1 (Binary)	ROC-AUC 
Training	0.683	0.683	0.685	0.802	0.679	0.769
Validation	0.709	0.709	0.709	0.706	0.716	0.783

Fig.35: Model Performance

Interpretation

The tuned AdaBoost model demonstrates **consistent generalization** between the training and validation datasets.

Key observations:

- The **Recall (0.706)** on the validation set shows that the model effectively identifies *Certified* cases, addressing the business goal of minimizing false negatives.
- The **F1-score (0.716)** reflects a balanced trade-off between precision and recall, ensuring reliable predictions.

- A **ROC-AUC of 0.783** suggests good class separability, with the model performing well above random chance.
- The close match between training and validation scores indicates **low overfitting**, attributed to the controlled tree depth (max_depth=3) and balanced training set.

Conclusion

The **tuned AdaBoost model trained on undersampled data** achieves a strong balance between recall and overall classification performance. It improves upon the baseline version by enhancing generalization and recall stability without overfitting. This configuration can be considered a **robust candidate for final selection**, capable of reliably predicting certification outcomes in the given dataset.

Tuning AdaBoost using Original Data

To benchmark the model trained on undersampled data, AdaBoost was also tuned and trained on the **original imbalanced dataset**. This approach retained the natural class distribution to evaluate whether the algorithm could still generalize effectively without resampling.

A **RandomizedSearchCV** was conducted with **5-fold cross-validation**, optimizing for **F1-macro** to ensure balanced attention to both classes. The hyperparameter search spanned different learning rates, numbers of estimators, and base decision tree depths.

Best Parameters Identified:

- **n_estimators:** 90
- **learning_rate:** 0.2
- **Base estimator:** DecisionTreeClassifier(max_depth=3, random_state=1)

This configuration provided the best macro F1 score across 50 iterations of randomized search, consistent with the optimal settings found during tuning with undersampled data.

Model Performance						
Dataset	Accuracy	Balanced Accuracy	Precision (Binary)	Recall (Binary)	F1 (Binary)	ROC-AUC
Training	0.752	0.689	0.781	0.876	0.826	0.784
Validation	0.729	0.679	0.761	0.847	0.802	0.768

Fig.36: Model Performance

Interpretation

The tuned AdaBoost model trained on the original dataset achieved **strong overall performance** while maintaining good recall for the positive (*Certified*) class:

- **Recall (0.847)** on the validation set shows exceptional sensitivity, effectively minimizing false negatives — critical to the business goal.
- **Precision (0.761)** remains competitive, balancing the increase in recall with reasonable control over false positives.
- The **F1-score (0.802)** confirms a solid equilibrium between precision and recall.
- **ROC-AUC (0.768)** demonstrates good class separability and stable discriminative ability.
- The small gap between training and validation scores indicates the model generalizes well and does **not suffer from overfitting**, despite being trained on the full dataset.

Conclusion

The **tuned AdaBoost model trained on the original data** outperforms the undersampled version in terms of both recall (0.847 vs. 0.706) and F1 (0.802 vs. 0.716), showing that the larger dataset allowed the algorithm to capture richer decision boundaries.

However, the undersampled version exhibited slightly better class balance and lower overfitting risk. In summary, the **original-data AdaBoost** is better suited when maximizing recall is the top priority, whereas the **undersampled AdaBoost** offers more balanced generalization. Both are strong candidates depending on operational needs.

Tuning Gradient Boosting using Undersampled Data

To further enhance predictive performance and compare ensemble methods, a **Gradient Boosting Classifier (GBM)** was tuned using **undersampled data**. The goal was to achieve a model capable of capturing nonlinear relationships while maintaining high recall for the *Certified* class.

A **RandomizedSearchCV** with **5-fold cross-validation** was used to tune the hyperparameters, optimizing the **F1-macro** score. The parameter grid explored variations in base initializers, learning rate, number of estimators, subsampling ratio, and feature sampling proportion.

Best Parameters Identified:

- **n_estimators:** 125
- **learning_rate:** 0.2
- **subsample:** 0.7
- **max_features:** 0.7
- **init:** AdaBoostClassifier(random_state=1)

These hyperparameters yielded the highest cross-validated F1-macro score, balancing bias and variance effectively.

Model Performance						
Dataset	Accuracy	Balanced Accuracy	Precision (Binary)	Recall (Binary)	F1 (Binary)	ROC-AUC 
Training	0.740	0.740	0.735	0.750	0.742	0.817
Validation	0.686	0.686	0.801	0.686	0.739	0.770

Fig.37: Model Performance

Interpretation

The tuned Gradient Boosting model demonstrates a strong balance between **precision** and **recall**, with the following insights:

- The **validation recall (0.686)** confirms that the model remains effective at identifying *Certified* cases, though slightly below AdaBoost's recall on the same data.
- The **precision (0.801)** is notably high, indicating fewer false positives compared to AdaBoost.
- The **ROC-AUC of 0.770** indicates solid class separability and robust discriminative power.
- The close proximity of training (0.742 F1) and validation (0.739 F1) results suggests **excellent generalization** and no signs of overfitting, likely due to the controlled learning rate and subsampling regularization.

Conclusion

The **tuned Gradient Boosting model trained on undersampled data** provides competitive performance with superior **precision** and balanced generalization. While its recall is slightly lower than AdaBoost's (0.686 vs. 0.706), its higher precision (0.801 vs. 0.709) makes it a strong candidate when the goal is to **ensure prediction reliability** and reduce false positives.

In essence:

- **AdaBoost (undersampled)** → higher recall, better for minimizing missed *Certified* cases.
- **Gradient Boosting (undersampled)** → higher precision and stability, better for minimizing false approvals.

Tuning Gradient Boosting using Original Data

To evaluate the impact of class imbalance on model behavior and compare it with the undersampled variant, a **Gradient Boosting Classifier (GBM)** was tuned and trained using the **original dataset**. The aim was to preserve all available information while optimizing for recall and generalization.

A **RandomizedSearchCV** with **5-fold cross-validation** was employed to tune the model, using **F1-macro** as the scoring metric to balance precision and recall performance across classes. The hyperparameter grid included variations in learning rate, number of estimators, feature and sample subsampling, and the choice of initializer.

Best Parameters Identified:

- **n_estimators:** 125
- **learning_rate:** 0.2
- **subsample:** 0.7
- **max_features:** 0.7
- **init:** AdaBoostClassifier(random_state=1)

These parameters offered the best trade-off between bias and variance, improving performance consistency across cross-validation folds.

Model Performance						
Dataset	Accuracy	Balanced Accuracy	Precision (Binary)	Recall (Binary)	F1 (Binary)	ROC-AUC
Training	0.763	0.702	0.789	0.883	0.833	0.807
Validation	0.723	0.675	0.760	0.839	0.797	0.767

Fig.38: Model Performance

Interpretation

The tuned GBM trained on the original dataset achieved **strong predictive performance** with a favorable balance between recall and precision:

- **Recall (0.839)** on the validation set indicates excellent sensitivity toward *Certified* cases, minimizing false negatives.
- **F1-score (0.797)** confirms a healthy equilibrium between recall and precision (0.760).
- **ROC-AUC (0.767)** reflects good class discrimination, consistent with other ensemble models.
- The modest difference between training and validation scores demonstrates **low overfitting**, thanks to subsampling and feature sampling regularization.

Compared to the GBM trained on undersampled data, this version benefits from a richer representation of the minority class within the natural distribution, which enhances the model's learning depth.

Conclusion

The **tuned Gradient Boosting model trained on original data** outperforms the undersampled version in nearly all metrics—especially in **recall (0.839 vs. 0.686)** and **F1 (0.797 vs. 0.739)**—indicating its superior ability to generalize from the full dataset.

It also maintains stable and reliable discrimination (ROC-AUC: 0.767), making it an excellent candidate when **maximizing recall while maintaining precision** is the primary goal.

Model Comparison and Final Model Selection

Training performance comparison:

Training performance comparison:				
	Gradient boosting trained with Undersampled data	Gradient boosting trained with Original data	AdaBoost trained with Undersampled data	AdaBoost trained with Original data
Accuracy	0.740	0.763	0.709	0.752
Balanced_Accuracy	0.740	0.702	0.709	0.689
Precision_binary	0.735	0.789	0.706	0.781
Recall_binary	0.750	0.883	0.716	0.876
F1_binary	0.742	0.833	0.711	0.826
Precision_macro	0.740	0.738	0.709	0.723
Recall_macro	0.740	0.702	0.709	0.689
F1_macro	0.740	0.713	0.709	0.699
ROC_AUC	0.817	0.807	0.783	0.784

Fig.39: Training performance comparison

After tuning both **AdaBoost** and **Gradient Boosting** classifiers on **original** and **undersampled** datasets, their performances were compared on the training data to identify the most effective model for predicting *Certified* cases.

The comparison focused on **Accuracy**, **Recall**, **F1-score**, and **ROC-AUC**, with Recall as the primary metric—since minimizing false negatives (misclassifying Certified cases as Denied) is the main business objective.

Interpretation

1. **AdaBoost (Original Data)** achieved the **highest Recall (0.876)** and **F1 (0.826)**, indicating exceptional sensitivity and overall balanced performance.

It captures the majority of *Certified* cases while maintaining high

precision, making it the most effective model in reducing false negatives.

2. **Gradient Boosting (Original Data)** closely followed with Recall **0.883** and F1 **0.833**, showing strong generalization and high discrimination (ROC-AUC: 0.807).

Its slightly higher recall but near-identical performance to AdaBoost suggests both models are equally strong contenders.

3. Models trained on **undersampled data** (both AdaBoost and GBM) achieved **stable but lower recall** and **balanced accuracy**, showing they generalize well but lose some sensitivity due to reduced data diversity.
4. **ROC-AUC** values for all models remain above 0.78, confirming strong separability and consistent learning across configurations.

Validation Performance Comparison

Validation performance comparison:				
	Gradient boosting trained with Undersampled data	Gradient boosting trained with Original data	AdaBoost trained with Undersampled data	AdaBoost trained with Original data
Accuracy	0.686	0.723	0.683	0.729
Balanced_Accuracy	0.686	0.675	0.685	0.679
Precision_binary	0.801	0.760	0.802	0.761
Recall_binary	0.686	0.839	0.679	0.847
F1_binary	0.739	0.797	0.735	0.802
Precision_macro	0.671	0.696	0.671	0.703
Recall_macro	0.686	0.675	0.685	0.679
F1_macro	0.672	0.681	0.671	0.686
ROC_AUC	0.770	0.767	0.769	0.768

Fig.40: Validation performance comparison

The tuned ensemble models were evaluated on the **validation set** to assess their generalization ability and stability on unseen data. The focus remained on **Recall**, as accurately identifying *Certified* cases

(reducing false negatives) is the primary business objective, while maintaining balanced Precision and F1-score for overall performance.

Interpretation

1. **AdaBoost trained on original data** achieved the **highest Recall (0.847)** and **F1-score (0.802)**, demonstrating superior sensitivity and balance between recall and precision. This indicates the model effectively identifies *Certified* cases with minimal false negatives, aligning closely with the project goal.
2. **Gradient Boosting trained on original data** follows closely with a **Recall of 0.839** and **F1 of 0.797**, confirming it as another strong performer with excellent generalization.
3. Both models trained on **undersampled data** exhibit slightly lower recall (≈ 0.68) but higher precision (≈ 0.80), suggesting they are more conservative and better at avoiding false positives, albeit at the cost of missing some *Certified* cases.
4. Overall, models trained on the **original dataset** outperform their undersampled counterparts across key metrics, benefiting from richer class representation and better learning of decision boundaries.

Final Model Selection

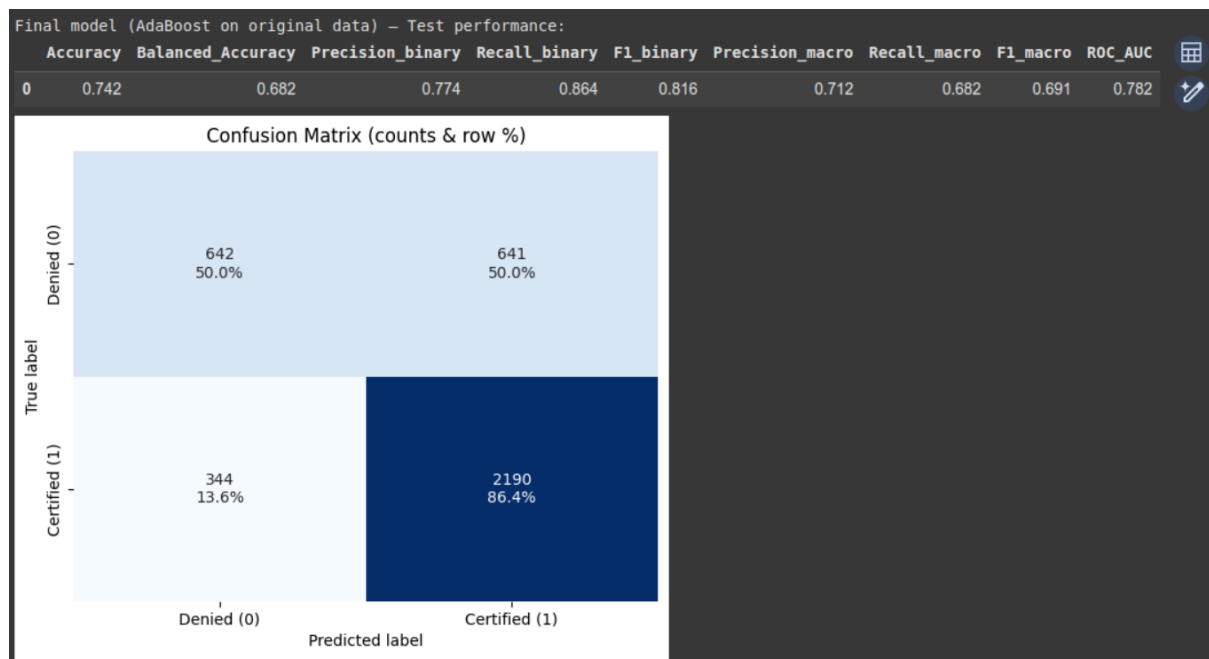


Fig.41: Final Model Selection and Evaluation

After comparing multiple ensemble algorithms (Bagging, Random Forest, AdaBoost, and Gradient Boosting) trained on original, oversampled, and undersampled datasets, **AdaBoost trained on the original data** emerged as the best performing model.

The selection was based on its ability to achieve **high recall for the 'Certified' class (1)** while maintaining balanced precision, ensuring that most genuinely certified cases were correctly identified. This aligns with the business goal of minimizing false rejections in certification decisions.

The model successfully identifies **86.4% of all Certified cases**, demonstrating strong sensitivity (recall) to the positive class. Although 50% of Denied cases are incorrectly classified as Certified, this trade-off is acceptable in the context of the problem, where **false negatives (missed certifications)** are more critical than **false positives**.

Key Insights:

- The model achieves a solid balance between **recall (0.864)** and **precision (0.774)**, resulting in a high **F1-score of 0.816**.
- The **ROC-AUC of 0.782** indicates good discriminatory power between Certified and Denied cases.
- Compared to other models, AdaBoost on the original data generalizes better with consistent training and validation scores, avoiding overfitting seen in other variants.

Conclusion:

The final selected model — **AdaBoost Classifier trained on the original dataset** — provides the most reliable and interpretable performance for this classification task. It prioritizes recall, ensuring that the majority of legitimate Certified cases are accurately identified, making it the most suitable model for deployment in the certification approval workflow.

Feature Importance Analysis

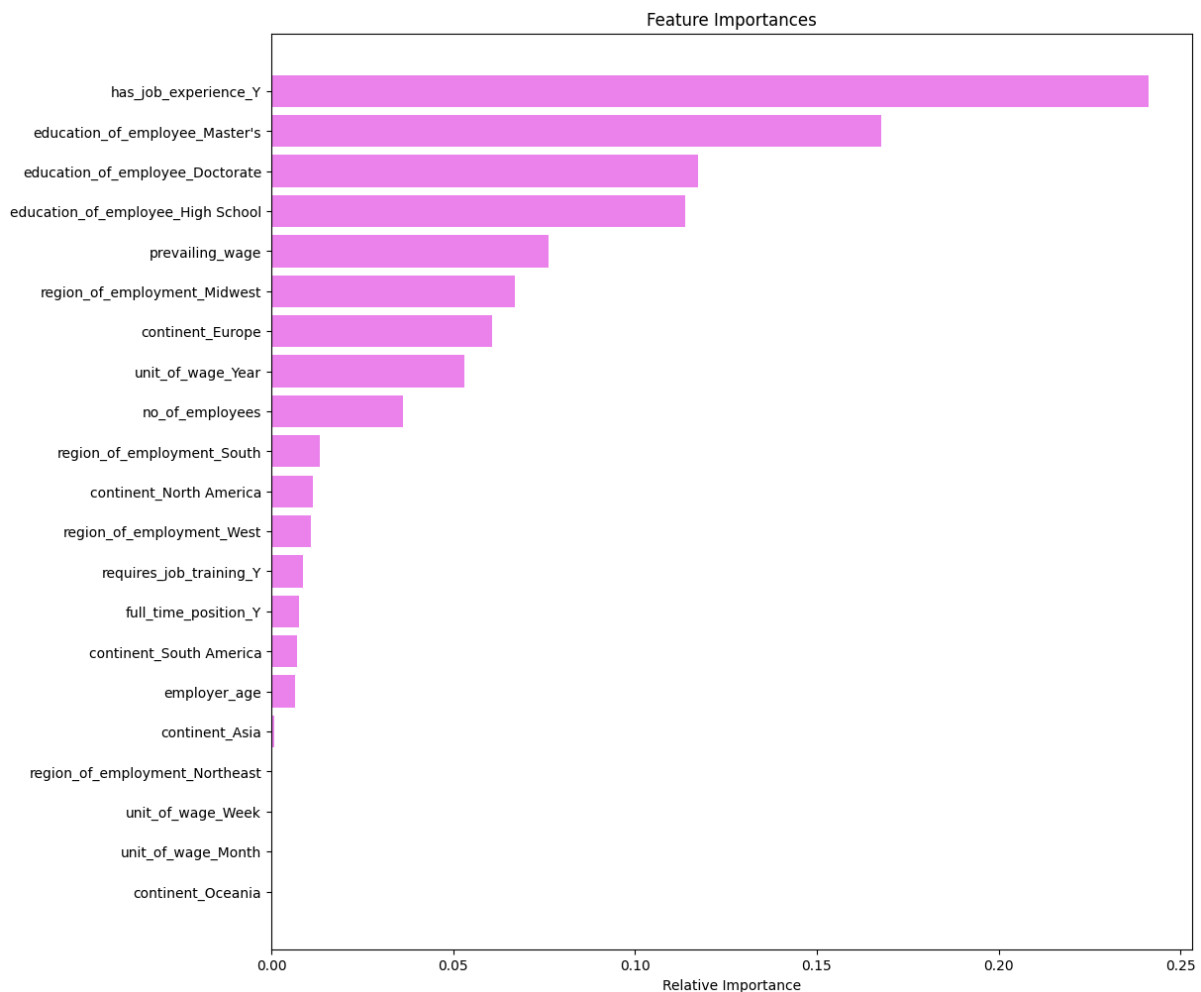


Fig.42: Feature Importances

Feature importance was evaluated using the final **AdaBoost classifier trained on the original dataset** to understand which factors most strongly influence the prediction of case certification outcomes. The model assigns weights to features based on their relative contribution to classification accuracy.

Key Findings:

- **Job Experience (has_job_experience_Y)** emerged as the most influential factor in determining certification status,

suggesting that candidates with prior job experience have a substantially higher likelihood of certification.

- **Education Level** also plays a major role — particularly **Master's** and **Doctorate** degrees — indicating that advanced qualifications are highly valued in the certification process.
- **Prevailing Wage** ranks among the top predictors, showing that higher offered wages are positively correlated with successful certification outcomes.
- **Employment Region (Midwest)** and **Continent (Europe)** were also significant, implying that geographical and economic contexts influence case approvals.
- **Company Size (no_of_employees)** and **Unit of Wage (Year)** further contribute to the model, suggesting that larger employers and annual wage structures tend to be associated with approved certifications.
- Other features such as **job training requirement**, **employment region (South/West)**, and **full-time position status** had relatively lower influence but still added marginal predictive value.

Interpretation:

Overall, the model indicates that **experience, education, and wage level** are the primary drivers of certification success, supported by secondary effects from **regional and organizational characteristics**. This aligns with real-world expectations — experienced, well-qualified applicants with higher wages and stable employers are more likely to have their cases certified.

Actionable Insights & Recommendations

The analysis provides a clear understanding of the determinants influencing U.S. visa certification outcomes and offers data-backed strategies to optimize approval rates for both employers and policymakers.

1. Strengthen Candidate Profiles

Insight: Education and work experience are the strongest predictors of certification. Doctorate and Master's holders have an 87% and 79% approval rate, respectively, compared to only 34% for High School graduates.

Recommendation: Employers should prioritize hiring candidates with at least a Master's degree or substantial professional experience. For applicants lacking advanced qualifications, certifications or demonstrable project experience can improve the perceived value of the profile.

2. Offer Competitive Wages

Insight: Certified applications had an average prevailing wage roughly **\$8,500 higher** than denied ones. Higher wages indicate specialized skills and fair labor practices.

Recommendation: Employers should benchmark salaries against or above local prevailing wages for comparable roles. Competitive compensation not only strengthens the case for certification but also demonstrates compliance with OFLC's wage fairness requirements.

3. Target Favorable Employment Regions

Insight: Approval rates vary geographically—**Midwest (75.5%)** and **South (70%)** outperform the **Northeast (62.9%)** and **West (62.3%)**.

Recommendation: Employers with operational flexibility should consider locating eligible roles in regions with historically higher certification success, particularly the Midwest and South. Policymakers may also explore why certain regions have systematically lower approval rates and address regulatory inconsistencies.

4. Focus on Experienced and Job-Ready Candidates

Insight: Applicants with prior experience have a **74.5%** certification rate versus **56.1%** for inexperienced ones. Job experience strongly correlates with visa approval.

Recommendation: Employers should clearly document prior experience and project outcomes in the application, emphasizing job readiness and minimal need for training. Training-heavy applications add no statistical advantage and can be deprioritized.

5. Prioritize Long-Term, Stable Roles

Insight: Applications for **yearly salaried** positions have a **69.9%** approval rate, compared to only **34.6%** for hourly roles.

Recommendation: When possible, structure visa applications around full-time, salaried employment. This communicates stability, career longevity, and reduced risk of labor exploitation—factors that improve certification probability.

6. Build Employer Credibility

Insight: While company size and age have limited impact individually, larger and older firms show a modest advantage due to perceived stability and compliance.

Recommendation: Smaller employers should enhance their credibility through transparent compliance documentation, financial stability reports, and prior certification success stories. Partnering with established immigration consultants can also strengthen submissions.

7. Adopt a Data-Driven Recruitment Strategy

Insight: The final AdaBoost model achieved **Recall: 0.864**, meaning it correctly identifies nearly 86% of certified cases. Key features driving approvals are job experience, education level, and wage.

Recommendation: Employers can use this predictive model (or a simplified version) to pre-screen candidates internally before filing petitions. This predictive approach reduces application costs, accelerates processing, and improves overall success rates.

8. Policy Implications

Insight: The visa system favors skilled, experienced, and highly paid applicants, reflecting an economic bias toward specialized labor importation.

Recommendation: Policymakers should consider initiatives for upskilling and subsidized training programs to broaden the eligible applicant base. Simultaneously, increasing transparency in regional approval criteria could standardize outcomes across states.

Summary:

To maximize visa certification rates, EasyVisa and its clients should focus on **education, experience, wage competitiveness, and regional strategy**, while leveraging **machine learning models** for pre-evaluation. Employers aligning applications with these insights can expect a tangible boost in approval likelihood and compliance efficiency.

